Assembly Instructions for the Yeah Hand™ Partly Completed Machinery

Manual for integration, configuration and use by integrators/assemblers

Translation of the Original Instructions

Company Name: VITTORIO-LUMARE

Product Name: VITTORIO-LUMARE Yeah Hand™

Model: Yeah Hand L / Yeah Hand R

Product Type: Robotic Hand

Product under development (prototype / partly completed machinery) intended solely for integrators and beta testers, not for end users.

Manufacturer's Address:

Vittorio Lumare

via Boccioni 4

88900 CROTONE (KR)

Italy

Revision history

Version	Comment	Author	Date
1.0	Initial version	Vittorio Lumare	20/09/2024
1.0 - IT	Italian translation	Fortunato Pantisano	11/11/2025
1.1 - IT	Translation completed (EU Declaration of Incorporation)	Vittorio Lumare	11/11/2025
1.2 - IT	EU Declaration of Incorporation and certificates updated	Vittorio Lumare	11/11/2025
1.3 - IT	Added chapter “Residual risks to be communicated to users”	Vittorio Lumare	12/01/2026
1.4 - IT	Assembly procedure updated	Vittorio Lumare	24/06/2026
1.5 - IT	Terms renamed, glossary terms added, sections on components no longer part of the product removed, illustration errors corrected	Vittorio Lumare	21/07/2026
1.6 – IT	Modified the thumb finger tendon-to-spool installation procedure	Vittorio Lumare	26/08/2026
1.7 – IT	Converted document from .odt to .docx for AI compatibility; many figures made static for positioning stability	Vittorio Lumare	27/08/2026
1.8 – IT	Added images of screws and nuts, added illustration of servo motor flange mounting, replaced thumb base illustrations, added photo showing the cables of the lowest servo motor, added CAD images of the components used, added photos of servo motors, tendons and Bowden tubes.	Vittorio Lumare, Oscar Bedolla (Public Invention)	05/10/2026
1.9 – IT	Added missing paragraph for M3 button-head screws; fixed missing links; corrected photo references in the paragraphs	Vittorio Lumare	14/11/2026

1 General notices and machinery conformity

1.1 Document structure

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1.2 How to use this manual

We invite readers to first read the safety chapter, which is the most critical and important aspect.

Afterwards, it is possible to skip to the desired chapter, each of which can be consulted independently, bearing in mind that each chapter refers to the others via links, so the section of interest can be consulted directly.

The final chapter contains the illustrations with the parts nomenclature, which should be consulted during the assembly procedure.

1.3 Legislative references

Directives	**Applicable requirements met**
Machinery Regulation (EU) 2023/1230	1.1.2, 1.1.3, 1.1.5, 1.3.2, 1.3.3, 1.3.4, 1.3.8.1, 1.5.1, 1.5.2, 1.5.5, 1.5.6, 1.6.1, 1.7.1, 1.7.2, 1.7.4, 1.7.4.1, 1.7.4.2, 4.1.2.3, 4.1.2.4, 4.3.1, Annex VI. It is declared that the relevant technical documentation has been compiled in accordance with Part B of Annex IV of the Machinery Regulation. The manufacturer undertakes to transmit, upon reasoned request from national authorities, the relevant technical documentation specified in Annex IV, Part B, within the required time limits.
2014/30/EU Electromagnetic Compatibility (EMC) Directive	Refer to the EMC Directive and the harmonised standards used below.
2011/65/EU Restriction of the use of certain hazardous substances (RoHS)	Refer to the RoHS Directive and the harmonised standards used below.

INTERNATIONAL STANDARD	**DESCRIPTION**
EN ISO 12100:2010	2012 Safety of machinery — General principles for design — Risk assessment and risk reduction
EN 60204-1	2009 Safety of machinery – Electrical equipment of machines – Part 1: General requirements
ISO 14539:2000	Manipulating industrial robots — Object handling with grasp-type grippers — Vocabulary and presentation of characteristics
ISO 14121-2	ISO/TR Safety of machinery – Risk assessment – Part 2: Practical guidance and examples of methods

1.4 Draft EU Declaration of Incorporation

Draft Original EU Declaration of Incorporation

Pursuant to Regulation (EU) 2023/1230, Annex V, Part B, of the European Parliament and of the Council, on machinery,

We, the manufacturer,

VITTORIO-LUMARE via Boccioni 4, 88900 CROTONE (KR) - Italy

We hereby declare that the following product:

Product Name: VITTORIO-LUMARE Yeah Hand

Description: Humanoid Robotic Hand

Functions: Object manipulation, Pick and Place, gestures.

Models: Yeah Hand L, Yeah Hand R

Types: Adaptive Electric Robotic Hand

Serial number: 2025 (anno) 00000 (Numerazione sequenziale, azzerata ogni nuovo anno)

The partly completed machinery which is the subject of this declaration must not be put into service until the machinery into which it is to be incorporated has been confirmed as being in conformity with the provisions of the Machinery Regulation. Conformity with all essential requirements of the Machinery Regulation depends on the specific robotic application and on the overall risk assessment.

Directives / Regulations	**Applicable requirements met**
EU Machinery Regulation 2023/1230	1.1.2, 1.1.3, 1.1.5, 1.3.2, 1.3.3, 1.3.4, 1.3.8.1, 1.5.1, 1.5.2, 1.5.5, 1.5.6, 1.6.1, 1.7.1, 1.7.2, 1.7.4, 1.7.4.1, 1.7.4.2, 4.1.2.3, 4.1.2.4, 4.3.1, Annex VI. It is declared that the relevant technical documentation has been compiled in accordance with Part B of Annex IV of the Machinery Regulation. The manufacturer undertakes to transmit, upon reasoned request from national authorities, the relevant technical documentation specified in Annex IV, Part B, within the required time limits.
2014/30/EU Electromagnetic Compatibility (EMC) Directive	Refer to the EMC Directive and the harmonised standards used below.
2011/65/EU Restriction of the use of certain hazardous substances (RoHS)	Refer to the RoHS Directive and the harmonised standards used below.

Person responsible for the documentation: Vittorio Lumare. Address: see manufacturer's address.

CROTONE, 07/09/2024 (place and date of issue)



Vittorio Lumare – Technical Director

VITTORIO-LUMARE

The following standards have been applied:

INTERNATIONAL STANDARD	**DESCRIPTION**
EN ISO 12100:2010	2012 Safety of machinery — General principles for design — Risk assessment and risk reduction
EN 60204-1	2009 Safety of machinery – Electrical equipment of machines – Part 1: General requirements
ISO 14539:2000	Manipulating industrial robots — Object handling with grasp-type grippers — Vocabulary and presentation of characteristics

ISO 14121-2	ISO/TR Safety of machinery – Risk assessment – Part 2: Practical guidance and examples of methods

1.5 Incorporated products

Incorporated products

- 5 Servo motors
  - Manufacturer: Shenzhen Waveshare Electronics Co., Ltd
  - Models: ST3215 Servo, ST3215-HS Servo Motor



Figure 1.1: WaveShare Servomotor ST3215-HS

- Control logic board
  - Manufacturer: Shenzhen Waveshare Electronics Co., Ltd
  - Model: Servo Driver with ESP32



Figure 1.2: WaveShare ESP32 Servo Driver

CERTIFICATE OF CONFORMITY

Certificate No.: CBT240825038EC

Certificate of Conformity

Applicant	:	**Shenzhen Waveshare Electronics Co., Ltd** 202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen
Manufacturer	:	**Shenzhen Waveshare Electronics Co., Ltd** 202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen
Product	:	**ST3215 Servo**
Model(s)	:	**ST3215 Servo** **ST3235 Servo,ST3020 Servo,SC09 Servo,WP90 Servo,** **ST3215-HS Servo Motor,SC15 Servo,ST3025 Servo,** **RSBL120-24,RSBL45-24,RSBL85-24,RSBL85-12**
Trademark	:	**WAVESHARE**
Test Standard	:	**EN 55032:2015+A1:2020** **EN 55035:2017+A11:2020** **EN IEC 61000-3-2:2019+A1:2021** **EN 61000-3-3:2013+A2:2021**

The Certificate of Conformity is based on an evaluation of a sample of the above mentioned products. It does not imply an assessment of the whole production. It is possible to use CE marking to demonstrate the compliance with this EMC Directive 2014/30/EU. It is only valid in connection with the test report number: CBT240825038ER.

CE

SHENZHEN CABLETEK COMPLIANCE LABORATORY LIMITED
CABLETEK
Manager
Aug. 26, 2024

Shenzhen CABLETEK Compliance Laboratory Limited
East 02, 6th Floor, Building 3, Zhouteng Industrial Zone, Shanglilang Community, Nanwan Street, Longgang District, Shenzhen

Tel: 400 666 1146 E-mail:cbt@cabletek.cn Web: www.cabletek.cn

CERTIFICATE OF CONFORMITY

Certificate No.: CBT241025045EC

Certificate of Conformity

Applicant	:	**Shenzhen Waveshare Electronics Co., Ltd.** 202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen
Manufacturer	:	**Shenzhen Waveshare Electronics Co., Ltd.** 202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen
Product	:	**Servo Driver with ESP32**
Model(s)	:	**Servo Driver with ESP32**
Trademark	:	**WAVESHARE**
Test Standard	:	**EN 55032:2015+A1:2020** **EN 55035:2017+A11:2020** **EN IEC 61000-3-2:2019+A1:2021** **EN 61000-3-3:2013+A2:2021**

The Certificate of Conformity is based on an evaluation of a sample of the above mentioned products. It does not imply an assessment of the whole production. It is possible to use CE marking to demonstrate the compliance with this EMC Directive 2014/30/EU. It is only valid in connection with the test report number: CBT241025045ER

SHENZHEN CABLETEK COMPLIANCE LABORATORY LIMITED *Approved* Manager

Oct. 30, 2024

Shenzhen CABLETEK Compliance Laboratory Limited
East 02, 6th Floor, Building 3, Zhouteng Industrial Zone, Shanglilang Community, Nanwan Street, Longgang District, Shenzhen

Tel: 400 666 1146 E-mail:cbt@cabletek.cn Web: www.cabletek.cn

1.6 Prototype / pre-series nameplate

The final nameplate, including any trademark, will be updated in the final version of the product.

TO DO INSERT LOGO
Manufacturer: VITTORIO-LUMARE

Model: YEAH-HAND-L / YEAH-HAND-R

TBD SN: 2026 000000
Year: 2026

2 Safety

2.1 Notices concerning hazardous situations

2.1.1 Meaning of symbols

This paragraph explains the meaning of the symbols used throughout the manual.

Please read and understand in this chapter the meaning of each symbol.



READ THE INSTRUCTION MANUAL



MANDATORY USE OF THE VICE



MANDATORY USE OF THE ASSEMBLY PLIERS



DO NOT SPRAY LIQUIDS ONTO THE ROBOTIC HAND



DANGER OF FINGER CUTS FROM CABLES



DANGER OF FALLING OBJECTS FROM HEIGHT



DANGER OF OBJECT EJECTION



DANGER OF FIRE



DANGER OF HOT SURFACES



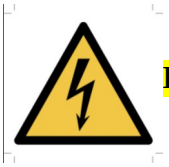
DANGER OF CUTS TO BODY PARTS



DANGER OF CRUSHING OF BODY PARTS



DANGER OF SUBSTANCES HARMFUL TO HEALTH



ELECTRICAL HAZARD

2.1.2 **Maintenance operations**

!! DO NOT PULL THE TENDONS WITH BARE HANDS !!





MANDATORY

- Use a vice to hold the part being worked on firmly in place
- Use assembly pliers to pull the tendons

POSSIBLE HARM:

- **Severe cuts to the fingers**
- **Blood loss requiring immediate surgery**

2.1.3 **Putting into service**

BEFORE THE PUTTING INTO SERVICE OF THE COMPLETE MACHINE THAT WILL INCORPORATE THE ROBOTIC HAND, THE FOLLOWING SYMBOLS MUST BE AFFIXED TO THE FINAL PRODUCT, SO THAT THEY ARE ALWAYS VISIBLE TO POSSIBLE PERSONS IN THE VICINITY



2.1.4 Typical operation

2.1.4.1 Persons near the robotic hand



WARNING: the robotic hand could break during use!

POSSIBLE HARM: if the robotic hand breaks, some broken parts could be sharp or pointed and could cut people's skin and cause permanent injury or blood loss and/or require immediate surgical intervention

2.1.4.2 Robotic hand holding an object



WARNING: an object could detach from the robotic hand and strike a person

POSSIBLE HARM: A thrown object could cause: breakage of bones, cuts or stab wounds, burns if very hot, contamination if it contains or is made of harmful substances.

2.1.4.3 Interaction with persons





WARNING: fingers or human skin can become trapped in the mechanisms and be injured

POSSIBLE HARM: a finger can be crushed to the point of causing severe bruising to skin or nails, a tendon can cut a human finger to the point of causing blood loss and/or require immediate surgical intervention, sharp parts can cut skin causing bleeding or bruising.

2.1.5 **Hazardous situations and residual risks**

Hazardous Situation	**Hazardous Event**	**Hazard**	**RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES**
Handling the tendon wires	Contact of human fingers with a tendon pulled under force	Cutting or abrasion of human fingers by the tendon wires	Skin injury if the operator does not follow the instructions during: • • removal/insertion of the tendon

2.1.6 **Permitted use of the machine and residual risks**

Hazardous Situation (ISO 12100:2010, Table B.3)	**Hazardous Event** (ISO 12100:2010, Table B.4)	**Hazard** (ISO 12100:2010, Table B.1)	**RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES**
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	The robotic hand breaks during operation, exposing sharp or abrasive parts	Impact, puncture, stabbing, abrasion	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • A broken part may be sharp and cut human skin, causing bleeding and requiring immediate surgery.
Being at a distance sufficient to be struck by an object held by the robotic hand, should the object detach from the robotic hand for any reason	Contact with an object ejected/thrown from the robotic hand	Impact, abrasion, cut, stabbing, puncture, burn, electrocution or contamination from hazardous substances caused by an ejected/thrown object	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • A thrown object may cause: bone fractures, cuts or cutting wounds if sharp, burns if hot, contamination if it contains or is made of harmful substances.
Intensive use of the robotic hand	Flame generated by overheated motors or the electronic board	Flame	On the complete machine, incorporating the robotic hand, manufactured by others, one of the following risks is possible: • The robotic hand may catch fire if used intensively beyond its limits.

2.1.7 **Non-permitted use of the machine and residual risks**

During permitted use of the robotic hand, once integrated into the final machine, the following hazardous situations may occur:

Hazardous Situation	**Hazardous Event**	**Hazard**	**RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES**
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	Contact with harmful parts of the held object: sharp edges, points, abrasive surfaces	Puncture, stabbing, abrasion, burn, electrocution	Puncture, stabbing, abrasion, burn, electrocution
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	Contact with a heavy object	Impact	Impact
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	Contact with the high-temperature part of the held object	Burns	Burns
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	Contact with the high-voltage part of the held object	Electrocution, burn	Electrocution, burn
Being near the robotic hand's working area (depends on the complete machine incorporating the robotic hand)	Contact with a held object contaminated by hazardous substances	Contamination by hazardous substances from part of the held object	Contamination by hazardous substances from part of the held object

It is important that integrators reduce the above risk situations, as they depend on the final machine and the associated risks cannot be reduced by acting on the robotic hand alone.

The following hazardous situations have been analysed and the related risks reduced by the manufacturer; they nonetheless still present residual risks.

Hazardous Situation (ISO 12100:2010, Table B.3)	**Hazardous Event** (ISO 12100:2010, Table B.4)	**Hazard** (ISO 12100:2010, Table B.1)	**RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES**
Touching the robotic hand's phalanges with human fingers	Contact of human fingers with moving parts of the robotic hand	Crushing of very small human fingers in the space between the robotic finger and the tendon, cutting of human fingers via forceful tendon traction	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • The finger may be compressed to the point of causing severe bruising to skin or nails • The tendon may cut the human finger to the point of causing bleeding and requiring surgery
Touching the palm of the robotic hand	Contact of human fingers with moving parts of the robotic hand	Crushing of human fingers or skin in the space between the thumb base and the palm or skeleton	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • The finger may be compressed to the point of causing severe bruising to skin or nails
Being under the robotic hand while it is	Object falling	Impact from a falling object	On the complete machine, incorporating the robotic hand,

holding an object			manufactured by others, one of these risks is possible: • A falling object may cause: bone fracture, cuts or stab wounds if sharp, burns if hot, contamination if contaminated
Touching an overheated robotic hand	Burn caused by contact with the flame generated by the robotic hand or by the overheated robotic hand	Burns	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • Skin burns caused by flames
Being at a sufficient distance from a damaged robotic hand to be struck by an object held by the damaged robotic hand, should the object detach from the damaged robotic hand for any reason	Contact with an object ejected/thrown from the damaged robotic hand	Impact, abrasion, cut, stabbing, puncture, burn, electrocution or contamination from hazardous substances by an ejected/thrown object	On the complete machine, incorporating the robotic hand, manufactured by others, one of these risks is possible: • A falling object may cause: bone fractures, cuts or cutting wounds if sharp, burns if hot, contamination if contaminated. • A thrown object may cause: bone fractures, cuts or cutting wounds if sharp, burns if hot, contamination if it contains or is made of harmful substances. • A broken part may be sharp and cut human skin, causing bleeding and requiring immediate surgery.

2.2 Personal protective equipment

2.2.1 Protective equipment for assembly and maintenance




During assembly and maintenance operations, it is mandatory to use the following tools:

- **Assembly pliers to hold and pull the tendons**
- **Vice to hold the robotic hand firmly in place during the operation**

2.2.2 Protective equipment for permitted use

During permitted use of the robotic hand, no special protective equipment is required.

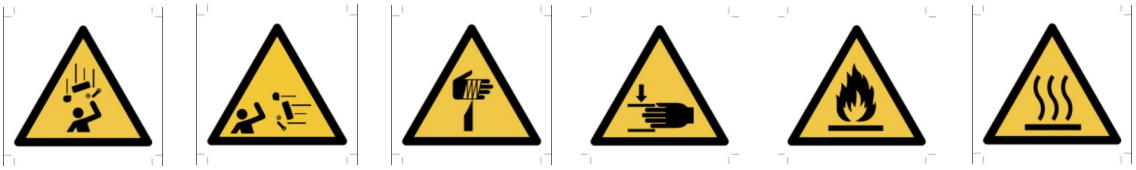
It is the responsibility of integrators to analyse further risk situations during use of the complete machine, and to add any appropriate protective equipment.

2.3 Warning plates affixed to the machine

These symbols must be affixed to the complete machine so that they are visible to the operators of the machine:



These symbols must be affixed to the complete machine before it is put into service, visible to anyone within the range of the complete machine to anyone, operators or simple passers-by or visitors:



This symbol must be affixed to the complete machine before it is put into service, visible to anyone within range of the complete machine, to anyone, operators, passers-by, or visitors, only where the complete machine incorporating the robotic hand is used to handle objects containing substances hazardous to health



2.4 Emergency situations

In case of fire, disconnect the power supply.

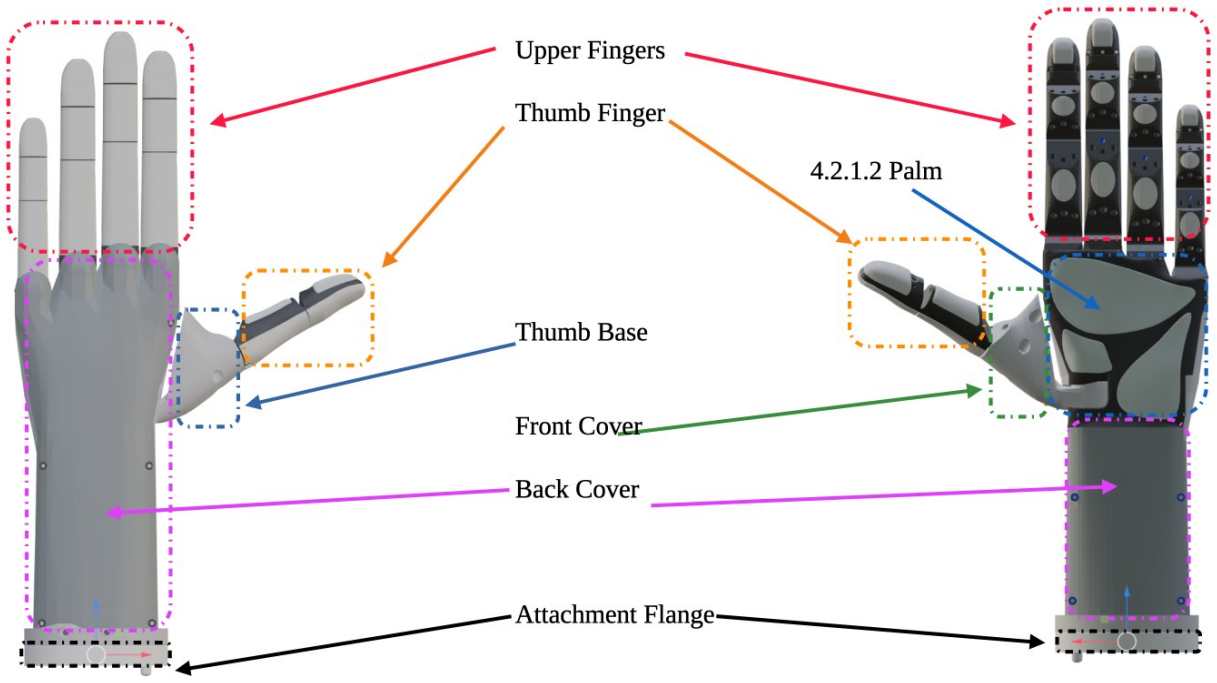
In case of contamination by hazardous substances, the integrators who manufactured the complete machine handling such substances, must carry out a risk analysis and assess emergency situations and record them in the technical file and in the instructions of the complete machine.

3 Machine overview

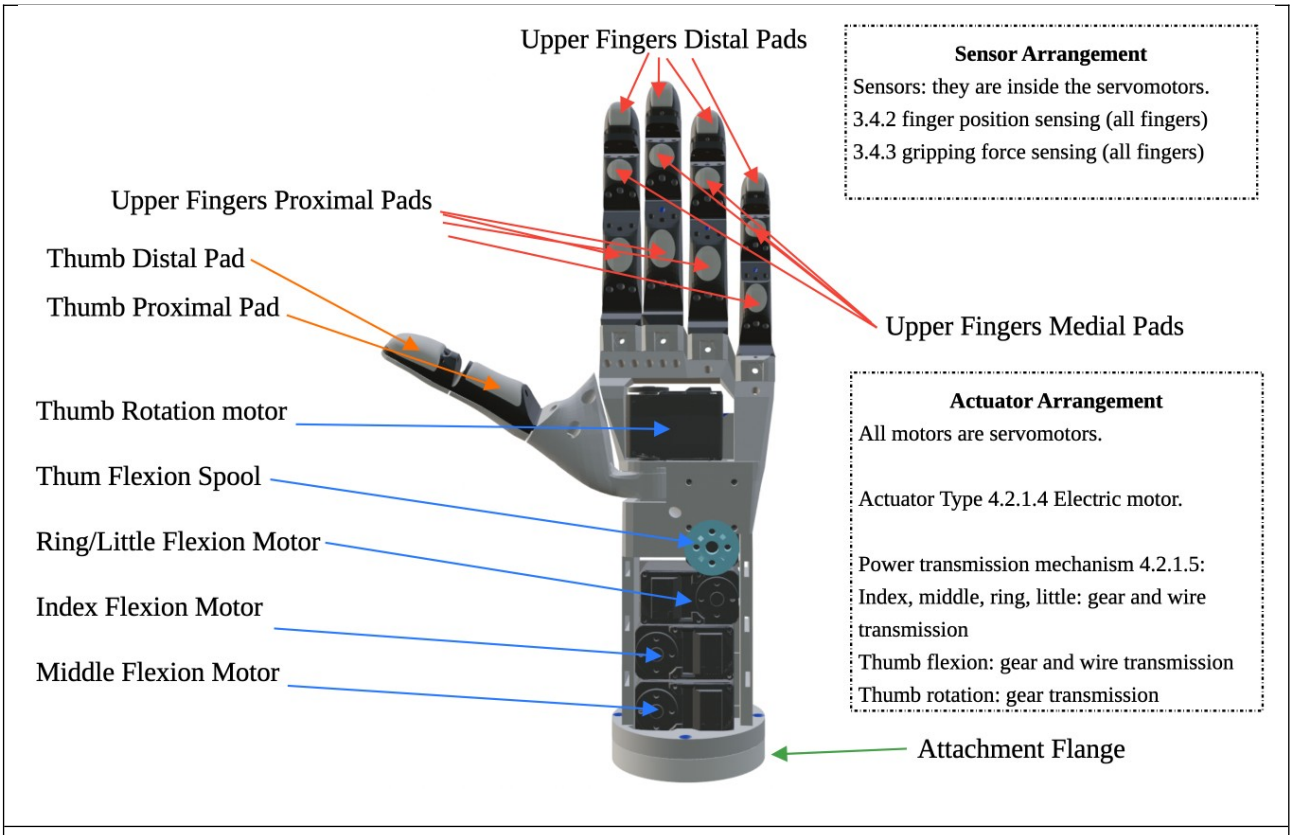
3.1 Machine description

Manufacturer	VITTORIO-LUMARE
Machinery Type	Robotic Hand (PARTLY COMPLETED MACHINERY)
Hand Model	Yeah Hand L (Left Hand), Yeah Hand R (Right Hand) The left-side model resembles the human "left" hand. The right-side model resembles the human "right" hand.
Type of robotic hand	Adaptive Electric Robotic Hand
Main applications of the robotic hand	Service robotics: • Grasping/releasing objects • Pick and place • Use of panel controls • Use of mechanical buttons • Hand gestures for human-robot interaction

Overall Technical Drawing - External View



Detailed Technical Drawing - Internal View



Sensor Arrangement
Sensors: they are inside the servomotors.
3.4.2 finger position sensing (all fingers)
3.4.3 gripping force sensing (all fingers)

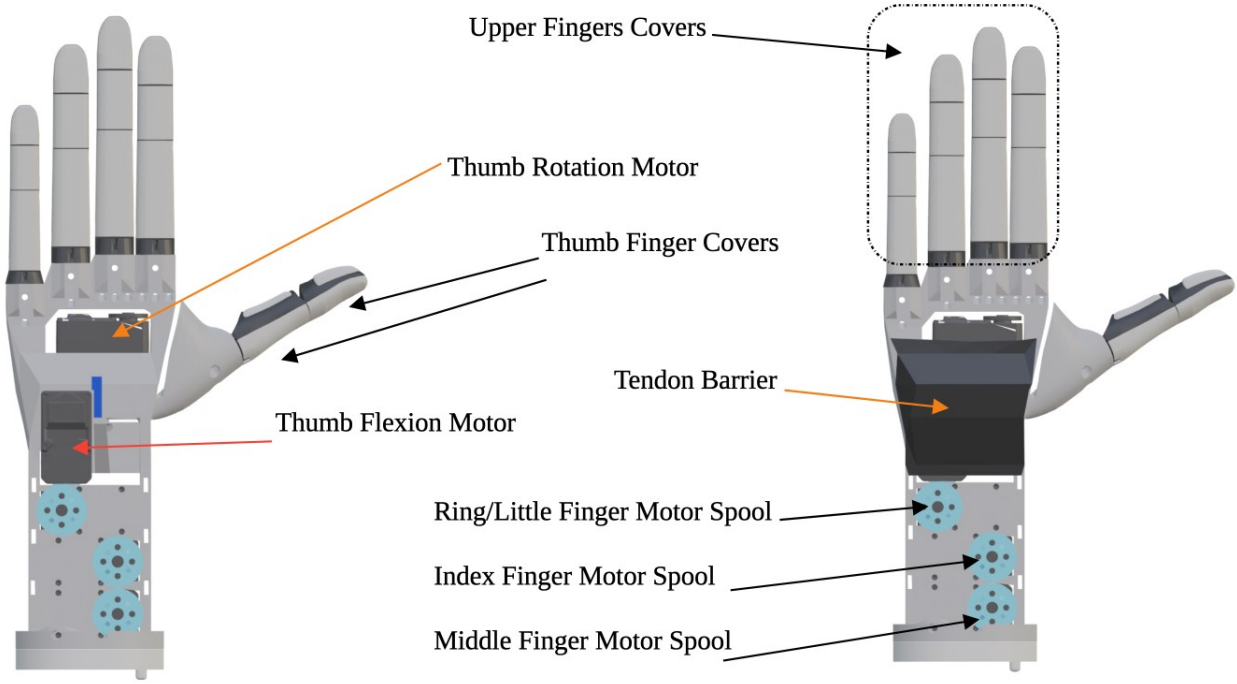
Actuator Arrangement
All motors are servomotors.

Actuator Type 4.2.1.4 Electric motor.

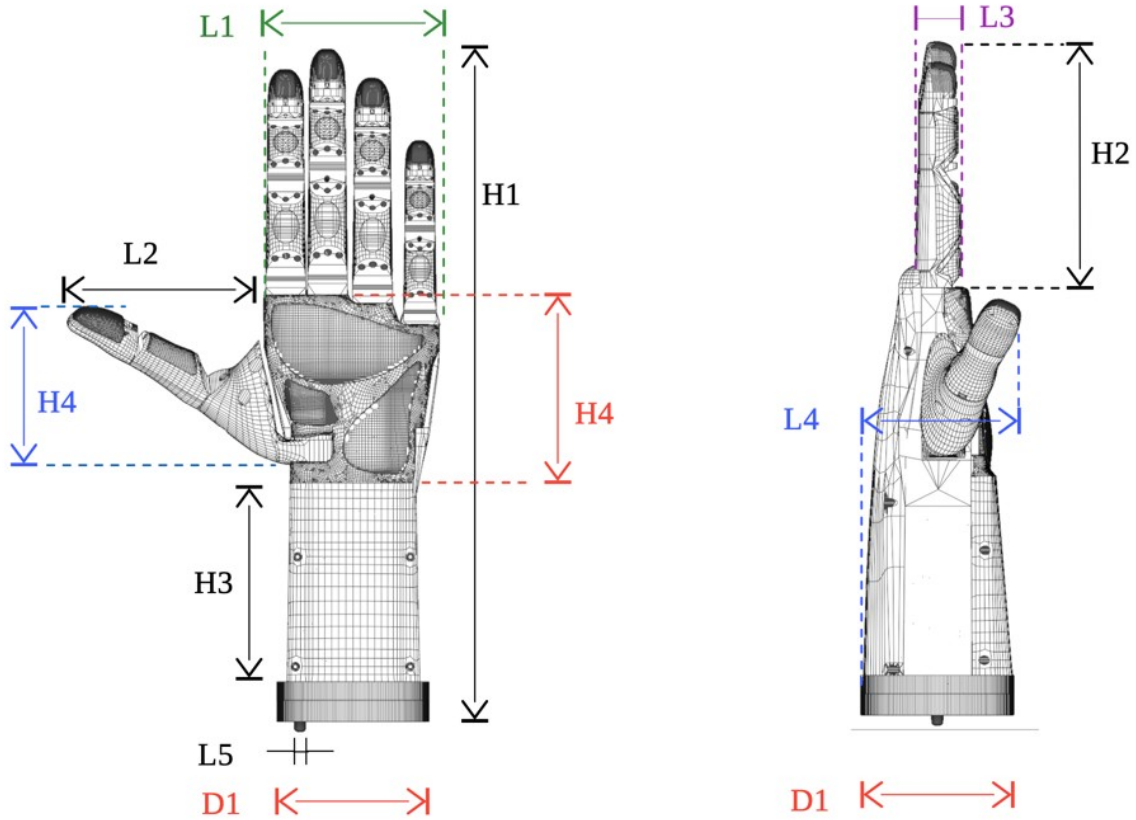
Power transmission mechanism 4.2.1.5:
Index, middle, ring, little: gear and wire transmission
Thumb flexion: gear and wire transmission
Thumb rotation: gear transmission

Actuator Arrangement
Thumb flexion motor:
Actuator Type 4.2.1.4 gear and wire transmission
Power Transmission mechanism 4.2.1.5 : gear and wire transmission

Sensor Arrangement
Sensors: they are inside the servomotors.
3.4.2 finger position sensing (all fingers)
3.4.3 gripping force sensing (all fingers)



Detailed Technical Drawing - Measures

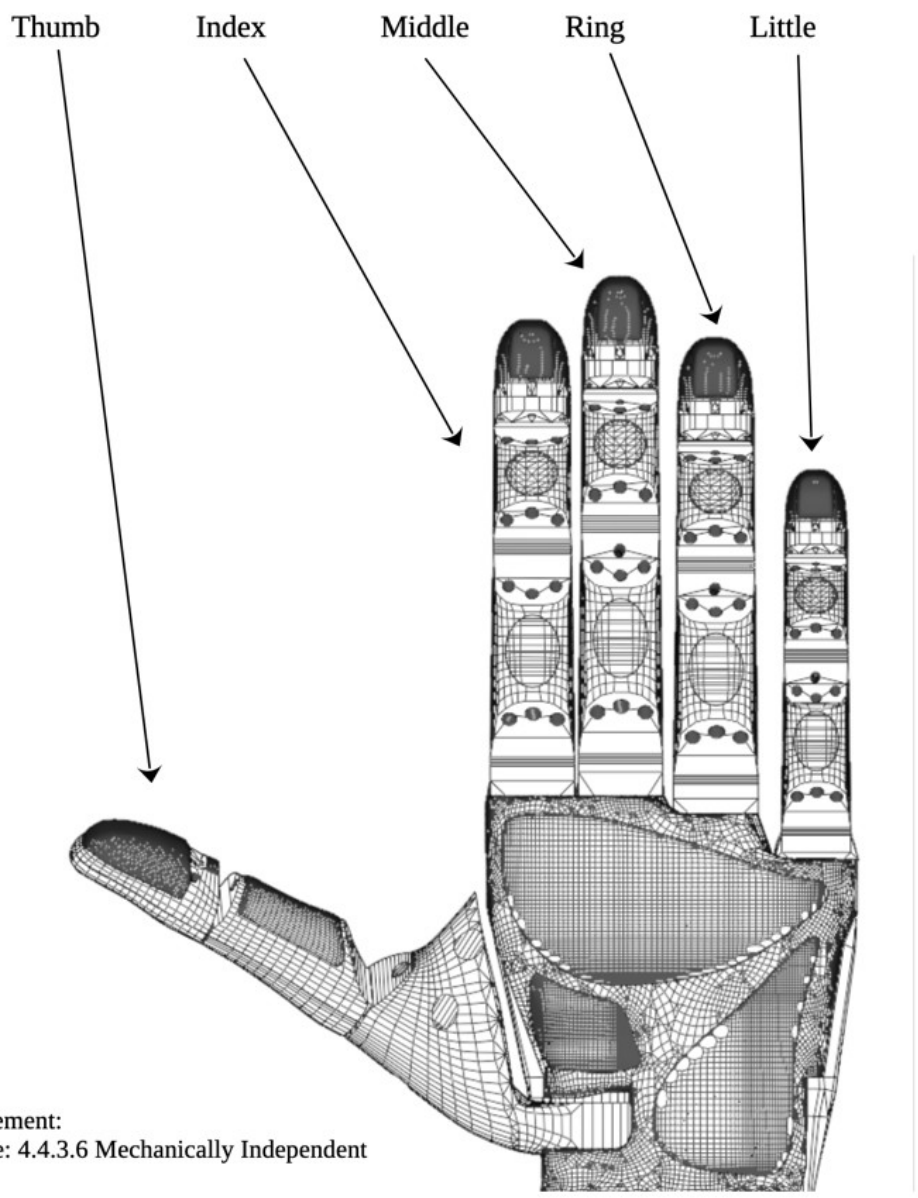


Measures
L1 : 83 mm
L2 : 99 mm
L3 : 21 mm
L4 : 74 mm
L5 : 5.5 mm
D1 : 71 mm
H1 : 318 mm
H2 : 117 mm

Tolerance : ±0.4 mm

Detailed Technical Drawing

Finger Arrangement



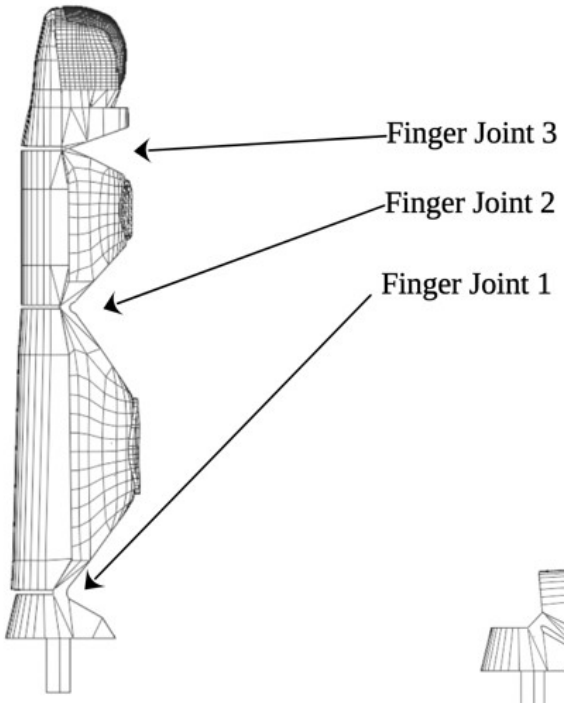
Type of Movement:
Index, Middle: 4.4.3.6 Mechanically Independent Fingers

Ring, Little: 4.4.3.5 Mechanically Interrelated Fingers (Ring and Little fingers move together)

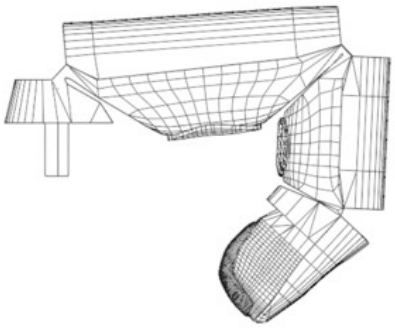
Detailed Technical Drawing

Upper Fingers Movement
It applies to Index, Middle, Ring and Little fingers

Flexion Angle: 70 degrees



Fully open finger

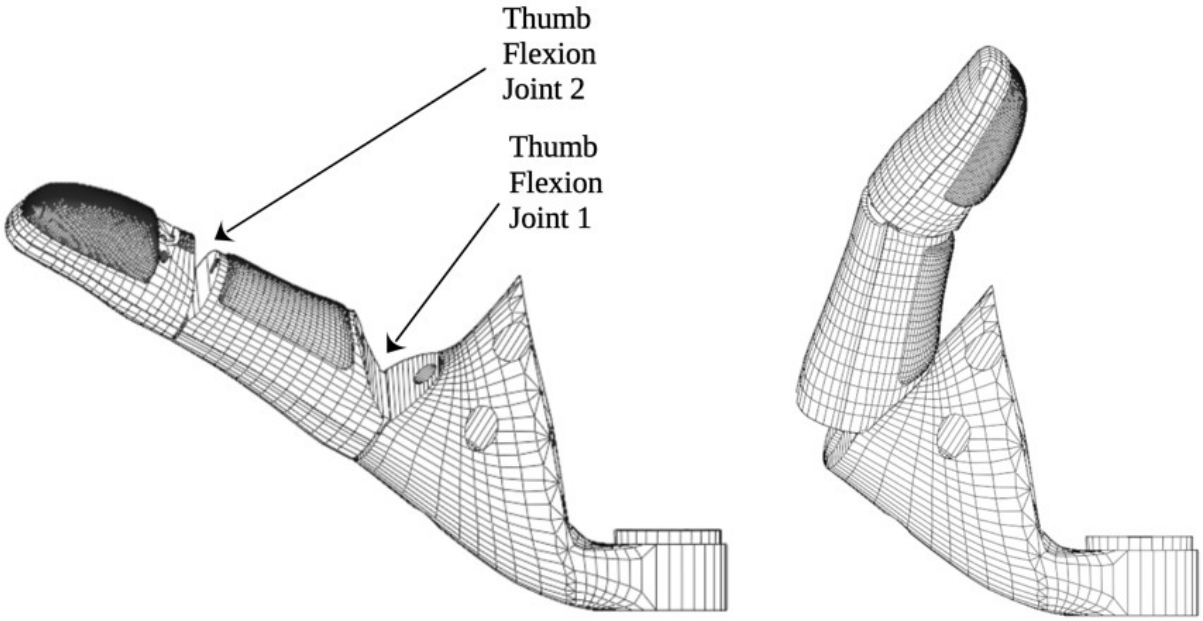


Fully closed finger

Type of finger movement: 4.4.3.3 Multi-Joint Finger
Flexion angles:
Finger Joint 1 : 87 degrees
Finger Joint 2 : 100 degrees
Finger Joint 3 : 43 degrees

Detailed Technical Drawing

Thumb Finger Movement : Flexion



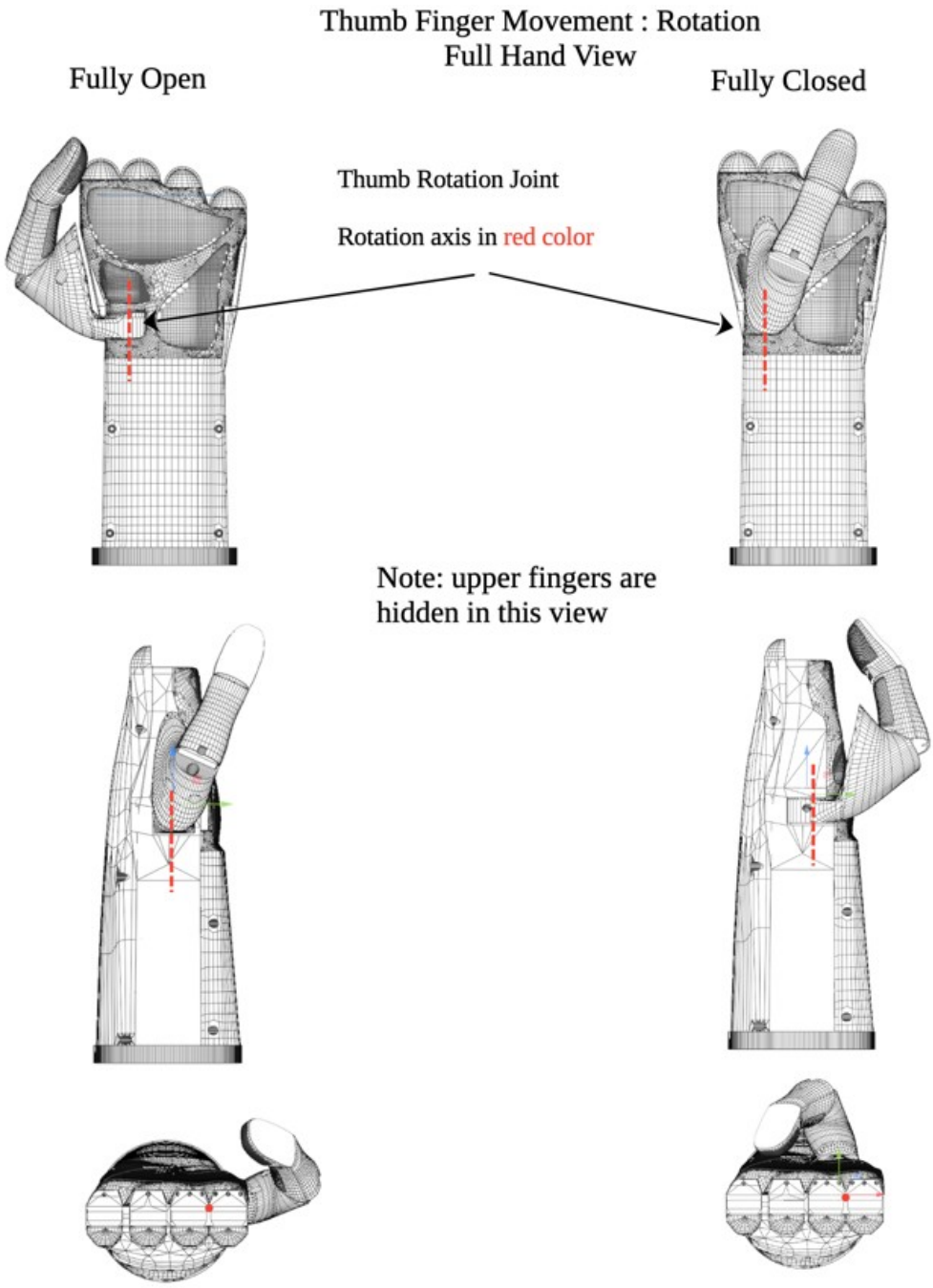
Fully open finger

Fully closed finger

Type Of Finger Movement:
4.4.3.3 Multi-Joint Finger
4.4.3.6 Mechanically Independent finger

Flexion Angles
Thumb Flexion Joint 1 : 70 degrees
Thumb Flexion Joint 2 : 25 degrees

Detailed Technical Drawing



Type Of Finger Movement: 4.4.3.3 Multi-Joint Finger
Rotation angle:
Thumb Rotation Joint : 79 degrees

3.2 Intended use

This electric robotic hand is adaptive and is designed for robotic applications.

The robotic hand must be installed on a handling device, such as:

- a mobile platform
- a robotic manipulator
- a humanoid service robot
- a non-humanoid service robot

The robotic hand is suitable for applications:

- educational:
  - schools
  - universities
- research:
  - laboratories
  - research centres
  - universities
- prototyping:
  - development of service robots (humanoid or otherwise) incorporating the robotic hand
- commercial
  - integrated into service robots:
    - in public areas
    - for domestic use
- hobbyist

The robotic hand is not suitable for industrial applications, as it does not possess sufficient:

- robustness
- repeatability
- high immunity to disturbances typical of industrial environments

3.3 Reasonably foreseeable misuse

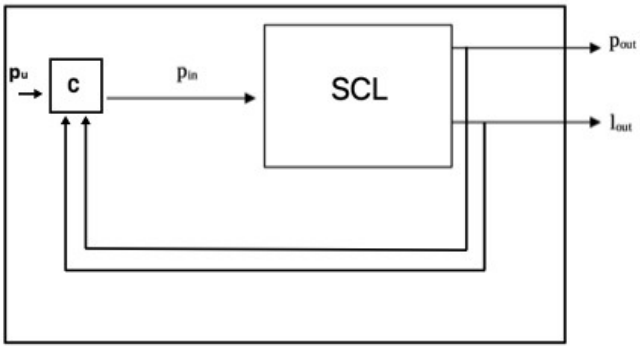
Misuse situations:

- coming into physical contact with living beings
- handling of skin or body parts
- shaking hands with persons or animals
- highly humid environments, or with water droplets or high-density vapours
- temperatures above 50 ˚C
- lifting objects heavier than 2kg
- handling substances hazardous to health, radioactive, or at high temperature (above 50 degrees)
- handling liquids or powdery substances, or with small particles (under 12 mm)

3.4 Technical characteristics of the machine

Robotic hand performance	
Mass of the robotic hand (kg)	650g without covers / 750g with covers
Maximum distance between fingers (mm)	160 mm
Minimum distance between fingers (mm)	0
Maximum grip force (N)	90N (TBD : calculated)
Maximum manipulation force (N)	35N (TBD : calculated)
Maximum allowable finger moment (N*m)	0.1
Grip surface (mm2)	1260
Operating time: Grasp (s)	2
Operating time: Release (s)	2
Type of robotic hand	
Number of fingers	5
Finger arrangement	Humanoid
Degrees of mobility of the hand (4.3.1)	Spatial grasp: 6
Degrees of freedom of the hand (4.3.2)	5
Type of grasp (4.3.6)	4.3.6.2 non-centric grasp 4.3.6.3 adaptive grasp 4.3.6.5 asymmetric grasp 4.3.6.6 grasp force
Fingers	
Degrees of mobility per finger (4.4.1)	Index: 3 Middle: 3 Ring: 3 Little: 3 Thumb: 5
Degrees of freedom per finger (4.4.2)	Index: 1 Middle: 1 Ring + Little: 1 Thumb: 2
Finger movement (4.4.3)	• Index: ◦ multi-articulated fingers (4.4.3.3) ◦ mechanically independent fingers (4.4.3.6) • Middle: ◦ multi-articulated fingers (4.4.3.3) ◦ mechanically independent fingers (4.4.3.6) • Ring: ◦ multi-articulated finger (4.4.3.3) ◦ mechanically interconnected finger (4.4.3.6) • Little: ◦ multi-articulated finger (4.4.3.3) ◦ mechanically interconnected finger (4.4.3.6) • Thumb:

	◦ multi-articulated fingers (4.4.3.3) ◦ mechanically independent fingers (4.4.3.6)
Actuators (4.2.1.4)	
Actuator type	Electric motors
Number of actuators	5
Rated actuator power (W)	60W
Power transmission mechanisms (4.2.1.5)	
Upper finger mechanism (index, middle, ring, little)	Gear + cable
Thumb flexion mechanism	Gear + cable
Thumb rotation mechanism	Gear
Note: "cable" refers to the tendon, a braided polyethylene fishing line	
Finger movement details (referenced to the mechanical interface coordinate system as defined in ISO 9787):	
TCP definition (3.3.8): offset from the XmYm plane	Zoffset = 146.1 mm
TCP definition (3.3.8): offset from the Zm axis	Xoffset = -0.1 mm Yoffset = 18.9 mm
Finger control	
Finger control scheme (4.5)	4.5.6 Hybrid Control
Control signals	Wireless serial character commands (BT, WiFi) • Signal 1: Select Mode ◦ Tendon Installation ◦ Tendon Calibration ◦ Relax (no movement) ◦ Grasp Control (POWER, POWERSMALL, MONKEY, PINCH) ◦ RAW Control (full control of each finger joint) • Signal 2: ◦ Fingers Closed/Open: 0-100%
Feedback signals	Wireless serial characters (BT, WiFi) • Position: integer value • Speed: integer value • Load: integer value • Temperature: integer value • Current: integer value • Movement: integer value (0: false, 1: true) • Status: integer value

Finger control block diagram


Sensing functions (3.4)	
Position sensing	Yes: Resolution: 4096, Range: [0, 360] degrees
Torque sensing	Yes: Resolution: 2000, Range: [-25, 25] Kg*cm
Temperature sensing	Yes: Accuracy: 1˚C, Range: [0, 70]˚C
Voltage sensing	Yes: Accuracy: 0.1V, Range: [0, 12]V
Current sensing	Yes (TBD)
Movement sensing	Value: 1: Moving, 0: Stationary
Clamping elements (4.6)	
Material	• Rigid polymer (e.g. ABS / PLA / PETG / nylon) • Flexible polymer (e.g. TPU / TPE) • Skin-safe silicone (e.g. Smooth-On Dragon Skin NV / Reschimica R PRO GLASS)
Replaceable	Yes
Interchangeable	Yes
Operating environment	
Temperature range	10 – 40 ˚C
Relative humidity range	0 – 80 %
IP protection	**IP20 (INSUFFICIENT, MUST BE >IP4X)**

pu : user desired position
pin : controller C commanded position
pout : real output position
lout : real output load
C : Gripper main controller
SCL: Servo motor Control Loop (inside the servo motor)

Object properties		
Geometry (shape, dimensions) (mm)	Shape: any shape Dimensions: maximum diameter 10 cm	
Object mass (kg)	0 - 2 kg **TBD**	
Max/min object temperature (˚C)	10˚C - 40˚C	
Surface characteristics	Anything free of: sharp edges/points, abrasive surfaces, harmful substances, liquids, radioactive materials	
Magnetic properties	< 1T	
Robotic interfaces (4.7) and wrist mechanisms:		
Mechanical interface	Compliant with standard	UNI EN ISO 9409-1:2005
	Designation code	ISO 9409-1-50-4-M6
Functionality (power and signal transmission)	Power transmission: direct current 12 V, max 14 A No signal transmission through the robot interface (transmission is wireless)	
Warning		
Risk (in the final machine) of: crushing, puncture, cutting, stabbing, fire, burn		

3.5 Controls and signalling

The robotic hand includes no physical controls; it is equipped with a wireless serial Bluetooth character interface.

Implementation of any controls is the responsibility of the integrators of the complete machine.

3.6 Environmental limits of the machine

Operating environment	
Temperature range	10 – 40 ˚C
Relative humidity range	0 - 80
IP protection	**TBD: IP40**

4 Transport, handling and storage

4.1 Transport

The robotic hand may be transported by hand.

It is recommended to transport the robotic hand in packaging that prevents:

- impact with surfaces that could compromise the structural integrity or the finish of the external surfaces.
- the ingress of liquids or dust, or of materials smaller than 12 mm TBD

It is recommended to remove the robotic hand from the machine on which it is integrated before transporting it, as it could be damaged during transport.

During transport, the temperature must not exceed 70 ˚C, otherwise geometric alterations of the materials it is made of may occur.

IMPORTANT: Before transporting the robotic hand, disconnect the power supply!

4.2 Handling

It is recommended to use a robotic manipulator to handle the robotic hand.

4.3 Storage

The robotic hand may be stored under the following conditions:

- temperature
  - 5˚C - 40˚C
- humidity :
  - 0 - 70% RH
- light:
  - store in the dark in a closed box
  - do not expose to UV rays

5 Assembly

These instructions explain how to assemble the robotic hand starting from its already manufactured components.

5.1 Servo motor installation and registration

This section describes how to install the servo motors in the hand's skeleton and how to register them on the logic board to ensure correct operation.

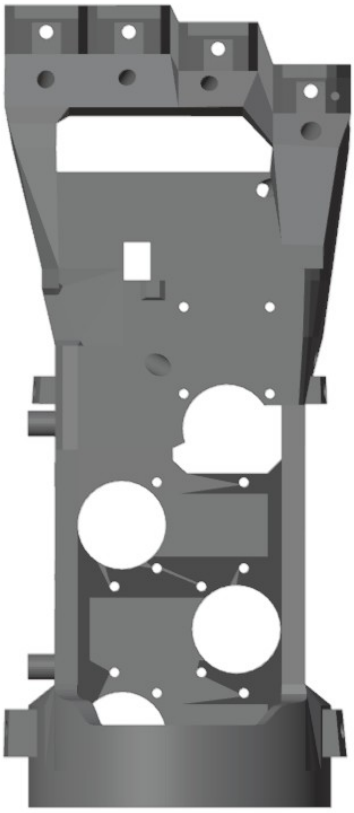


Figure 5.2: Left Hand - Skeleton - Front View – no components mounted

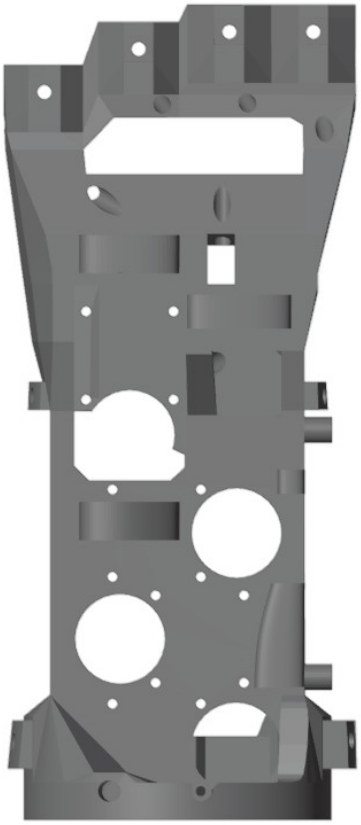


Figure 5.1: Left Hand - Skeleton - Rear View – no components mounted

5.1.1 **Installing the flange on the servo motor**

Before installing the servo motors on the hand skeleton, the flange must be installed on each servo motor.

The circular aluminium flange must be fitted onto the motor shaft and secured with the screw provided.

Note: Each servo motor is supplied with all its screws and the required flange. The flange to be mounted is the one with the toothed seat, which fits the motor shaft perfectly. The flange on the other side must not be mounted.

See Figure 5.3.

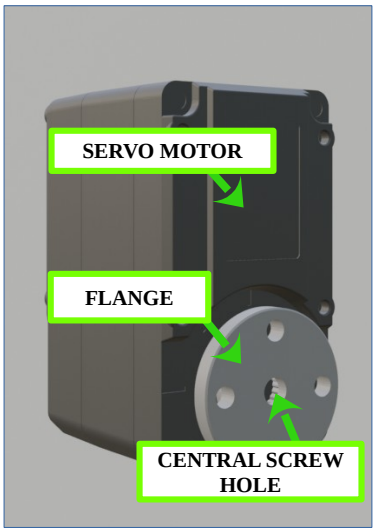


Figure 5.3: Installing the Flange on the Servo Motor

5.1.2 **Installing the Thumb Base and Thumb Rotation Servo motor (20 min)**

Note: TRS is an acronym for Thumb Rotation Servo
Installation:

1. Install the Thumb Base on the TRS flange (4 M3 countersunk-head screws with hex socket, 16 mm long) as shown in figure 5.4.
2. Install the TRS on the TRS Platform as shown in figure 5.5 and fasten it with 4 M2 self-tapping screws, button head, Phillips drive, max 6.5 mm long (do NOT use longer screws, otherwise the servo motor will be damaged).
3. Insert the TRS Platform + Thumb Base assembly onto the skeleton as shown in figure 5.6 and fasten it with 4 M2 cylindrical-head screws with hex socket, 5 mm long.

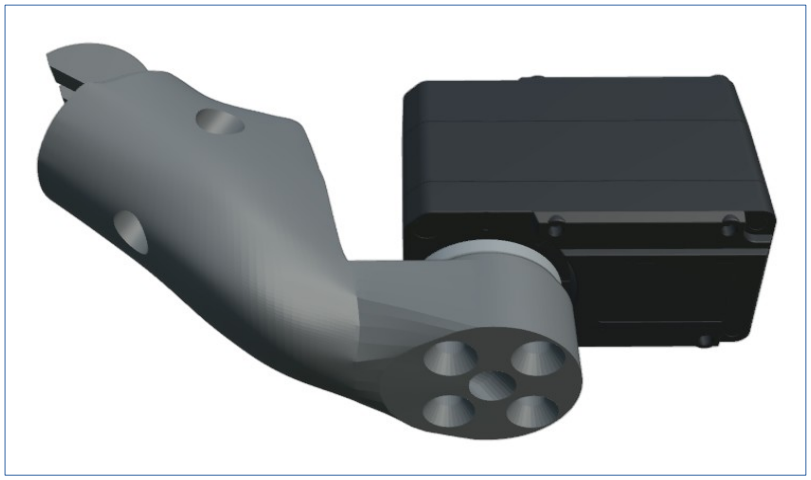


Figure 5.4: Thumb Base on the Thumb Rotation Servomotor

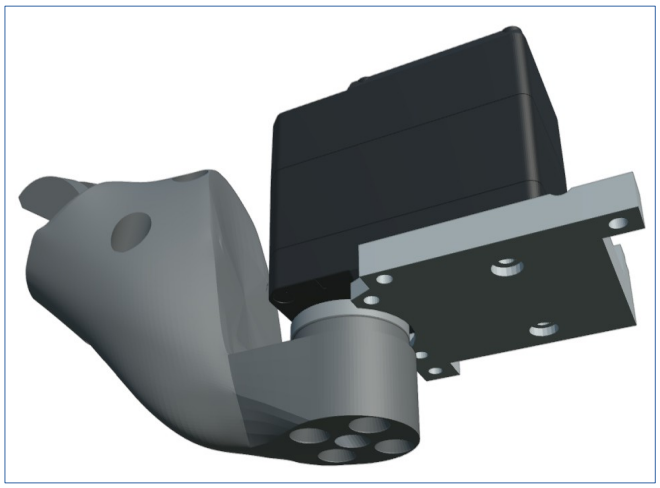


Figure 5.5: Thumb Base and TRS on TRS Platform

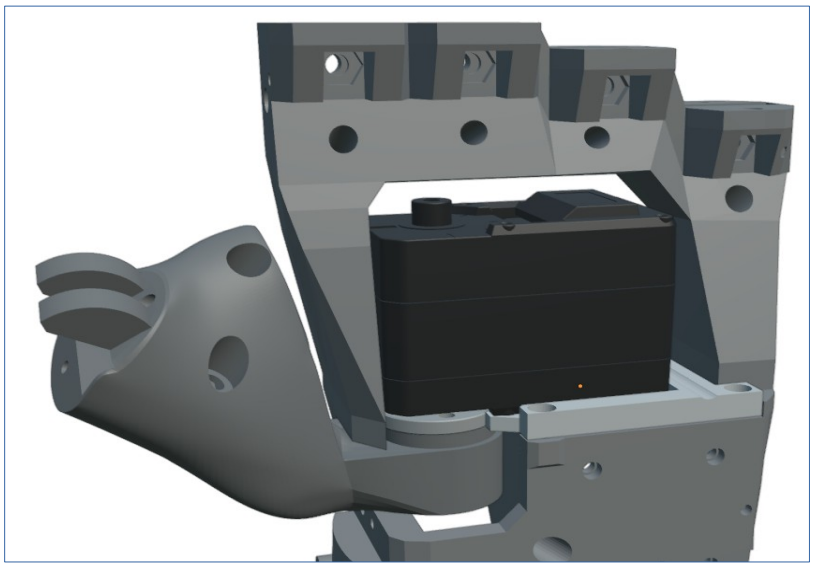


Figure 5.6: TRS and TRS Platform installed on the skeleton

5.1.3 **Installing the 4 finger-flexion servo motors (15 min)**

Note: each flexion servo motor (there are 4) must be fastened in its slot using 4 M2 self-tapping screws, 6.5 mm long.

1. Install each servo motor in its slot as shown in figures 5.7 and 5.8
2. Screw the servo motor to the skeleton holes.
3. **WARNING: Use the screwdriver provided and tighten the screws lightly to avoid damaging the servo motor's threaded seat, which is very delicate.**
    1. Note for the lowest servo motor (the one at the base of the skeleton):
        1. remove the central screw of the rotating flange to allow insertion into the slot. (see figure 5.3, which shows the central screw)
        2. connect 2 control cables to the servo motor, as it will not be possible to insert them afterwards.
        3. Insert the servo motor into the slot
        4. **Re-tighten the central screw on the flange.**
4. Make sure that all servo motor control cables are disconnected except those of the lowest servo motor. The cables of the lowest servo motor must be routed out from above, as shown in figure 5.9

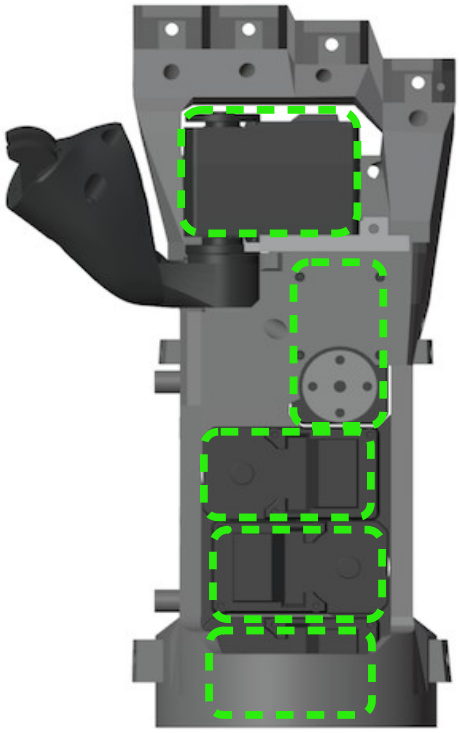


Figure 5.7: Flexion servo motors installed - front view

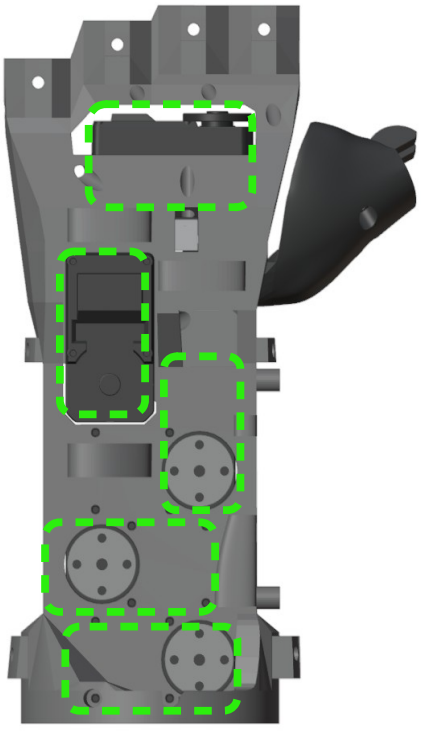


Figure 5.8: Flexion servo motors installed - rear view

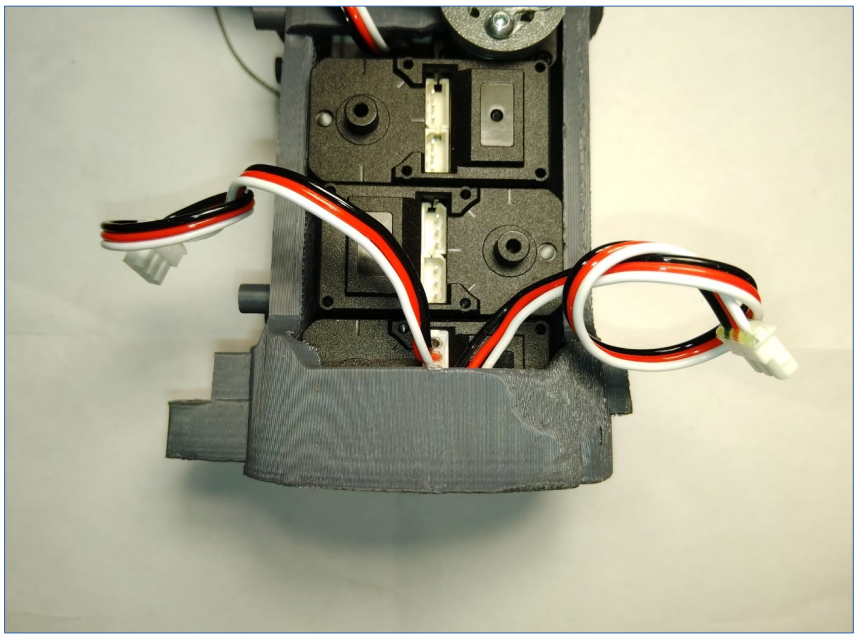


Figure 5.9: The cables connected to the lowest servo motor must exit from above

5.1.4 Servo motor registration and calibration

Before starting:

- Make sure the logic board is
  - removed from the hand
  - powered off and disconnected from both the power supply and the USB cable.
- Disconnect all servo motors from the logic board.
- If you use Windows: install the CP210x USB driver
- Open Arduino IDE
- Open Board Manager: install Espressif ESP32 version 3.3.10 or higher.
- Select the Board: ESP32-WROOM-DA Module
- Open Library Manager:
  - install SCServo version 1.0.2 or higher
  - install BluetoothSerial version 1.1.0 or higher
- Load the ***UniversalFlexionServoRegistration*** program in Arduino IDE and upload it to the logic board with a USB cable.
  - Download website: https://github.com/opsobot/yeah-hand/tree/main/firmware/src/Servo Registration
- **Disconnect the USB cable from the PC!**
- Power the logic board through the power connector (not through the USB cable!) using an external 12 V DC, 5 A power supply, connected to the circular connector as shown in figure 5.14
- Connect to the board via Bluetooth using a computer or a smartphone with a terminal program.
  - It is recommended to use a phone with the Serial Bluetooth Terminal app
  - The board appears as 'Yeah Hand' among the Bluetooth devices.

Important: never power the board with the USB cable connected and the external jack power supply at the same time!

Procedure (for each servo motor):

1. If the servo motor is a flexion servo (finger bending via tendons):
   1. Screw a fastening bolt into one of the holes of the circular plate.
      1. M3 button-head bolt with hex socket, 10 mm long
   2. Rotate the plate until the hole with the bolt is at the bottom right (figure 5.11).
2. If the servo motor is the thumb rotation servo:
   1. Position the thumb at approximately the middle of the range (figure 5.10).
3. Connect the servo motor to the logic board (leave all the other servo motors disconnected).
4. The program will ask for
   1. Servo motor ID: enter the appropriate ID from the Bluetooth terminal and send the command
      - Index servo motor: ID 1
      - Middle servo motor: ID 2
      - Ring/little servo motor: ID 3
      - Thumb servo motor: ID 4
      - Thumb Rotation servo motor: ID 5
5. The program will register the servo motor ID and the central position of the range
   1. At the end of the procedure the servo motor will make 4 oscillations of the plate (to confirm a successful result)

Once all the servo motors have been completed:

- Switch off the logic board by disconnecting the power supply.
- Disconnect the servo motor from the logic board.

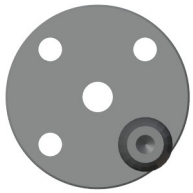


Figure 5.11: Flexion servo motor registration

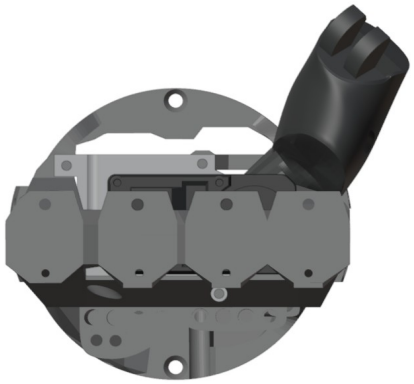


Figure 5.10: Thumb rotation servo motor registration

5.2 Programming the logic board with the operating program

Once the servo motors have been registered, the board must be programmed with the robotic hand's operating program. The program is 'RebeliaRobot'.

1. DISCONNECT THE SERVO MOTORS FROM THE BOARD
2. Open Arduino IDE
3. Select the board: ESP32-WROOM-DA Module
4. Open the RebeliaRobot.ino program
5. Disconnect the power connector from the logic board
6. Connect the logic board to the PC with the USB cable
7. Upload the program from Arduino IDE
8. Wait for the operation to complete
9. Disconnect the USB cable from the logic board
10. Connect the power connector to the logic board

5.2.1 **Servo motor connection**

The servo motors are connected to the logic board in a daisy chain, see figures 5.12 and 5.13.

1. Connect the logic board to servo motor 2
2. Connect servo motor 2 to servo motor 3
3. Connect servo motor 3 to servo motor 1
4. Connect servo motor 1 to servo motor 4, first passing the cable into cavity S1 and out through cavity S2
5. Connect servo motor 4 to servo motor 5, passing the cable into cavity S3

At this point, all servo motors are connected to the logic board in a daisy chain.

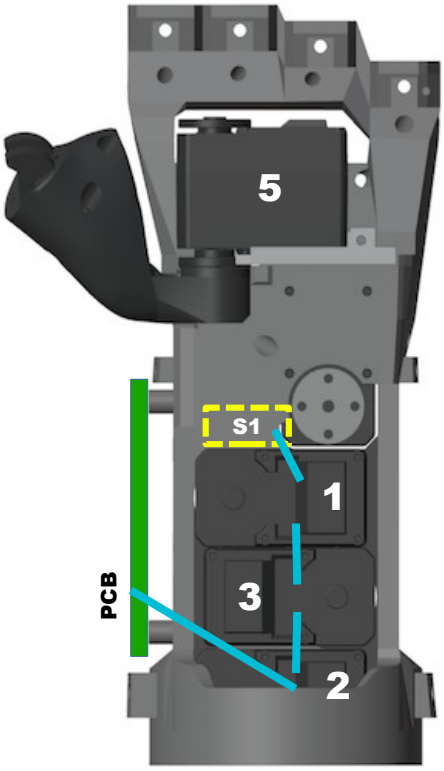


Figure 5.12: Front view. ID servo motors, logic board PCB and cavity S15.12

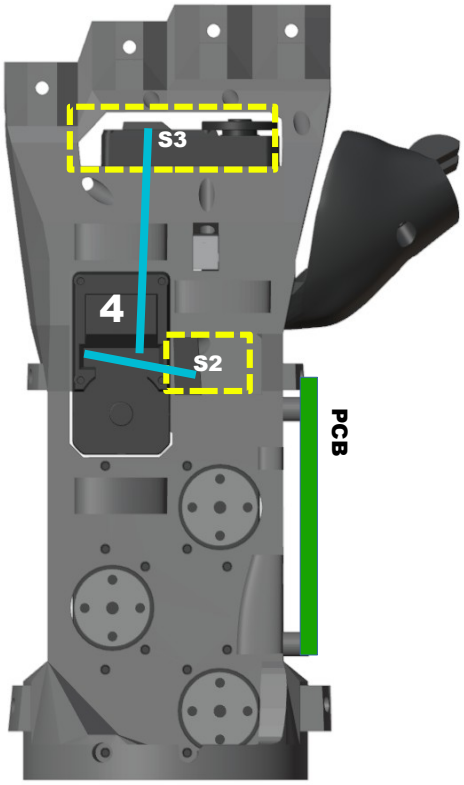


Figure 5.13: Rear view. ID servo motors, logic board PCB and cavity S2

5.3 Installing the logic board

1. Install the logic board on the skeleton as shown in figure 5.15, using 4 M2 self-tapping screws, 6.5 mm long.

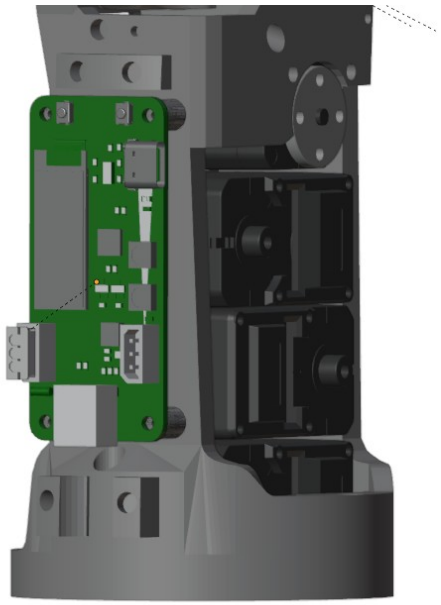


Figure 5.15: Logic board installed

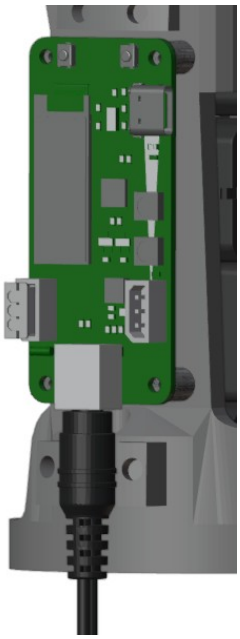


Figure 5.14: Power connector inserted, 12V DC, 5A

5.4 Installing the tendons in the fingers

5.4.1 Installing tendons in the upper fingers



WARNING: the tendons can cut fingers!

MANDATORY: use the vice to secure the skeleton, and the assembly pliers to pull the tendon.

Installing the tendon for the index finger (the same procedure also applies to the middle, ring and little fingers):

Note: at present, the assembly kit is supplied with the tendons already installed on the fingers.

5.4.2 Installing the tendon in the thumb



WARNING: the tendons can cut fingers!

MANDATORY: use the vice to secure the skeleton, and the assembly pliers to pull the tendon.

Note: at present, the assembly kit is supplied with the tendons already installed on the fingers.

5.5 Installing the Index, Middle, Ring and Little fingers

Note: each finger must be fastened, in its slot on the skeleton, with one M3 countersunk-head screw with hex socket, 12 mm long.

1. Insert the finger into its slot on the skeleton (figure 5.16)
    1. leave the tendons out; they will be inserted into the skeleton holes later, in another procedure
2. Position the M3 hex nut in its seat on the front of the skeleton (figure 5.17)
3. Secure the nut with the provided insertion tool (figure 5.19)
    1. Note: the tool is also used to hold the nut steady in its hexagonal seat while the bolt is being tightened
4. Insert the bolt into the hole on the rear of the skeleton and tighten it (figure 5.18)

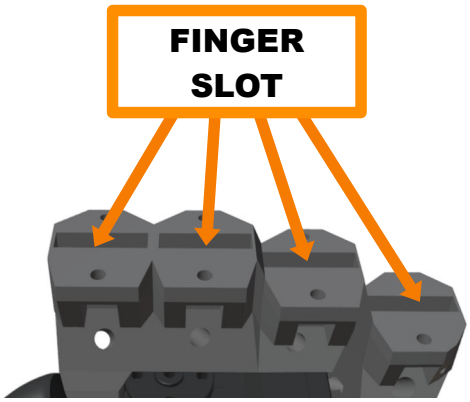


Figure 5.16: Finger slot

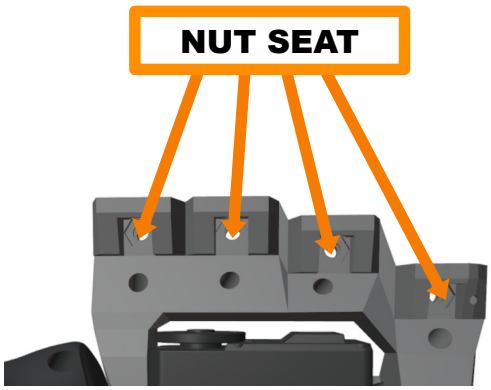


Figure 5.17: Nut seat

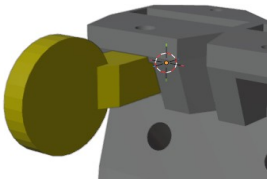


Figure 5.19: Nut insertion and holding tool

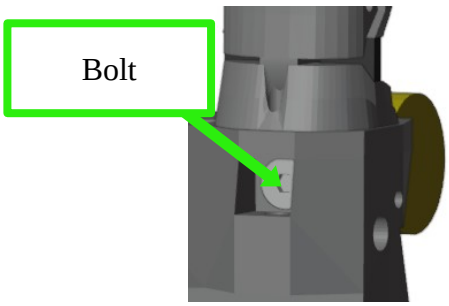


Figure 5.18: Finger bolt insertion

5.6 Installing the Thumb finger

Note: the thumb must be fastened, in its slot on the thumb base, with two M3 countersunk-head screws with hex socket, one 16 mm long and one 20 mm long.

1. Insert the finger into the slot (figure 5.20)
2. Insert two screws from behind (figure 5.20)
    1. Screw 1: M3, 16 mm long
    2. Screw 2: M3, 20 mm long
3. Secure with two M3 nuts placed on the opposite side (figure 5.21)

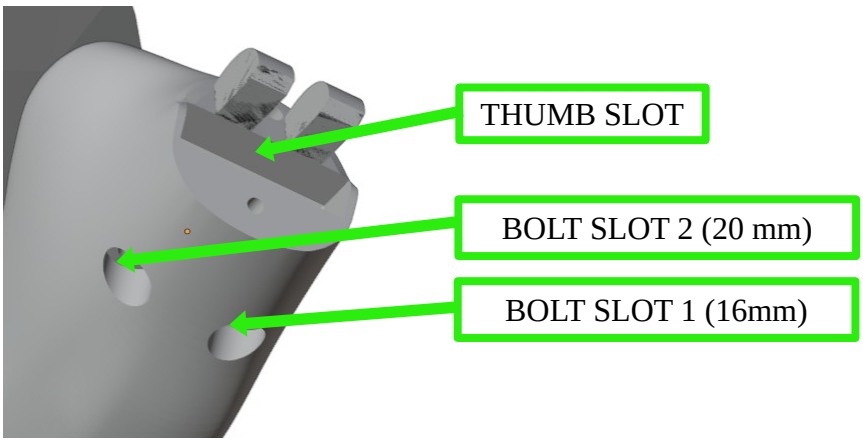


Figure 5.20: Thumb slot and bolt slots

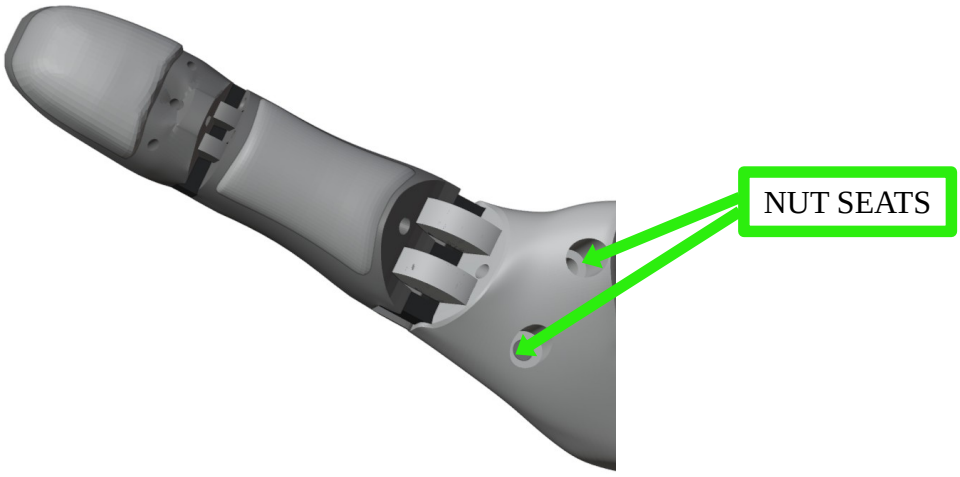


Figure 5.21: Thumb fastening with nuts

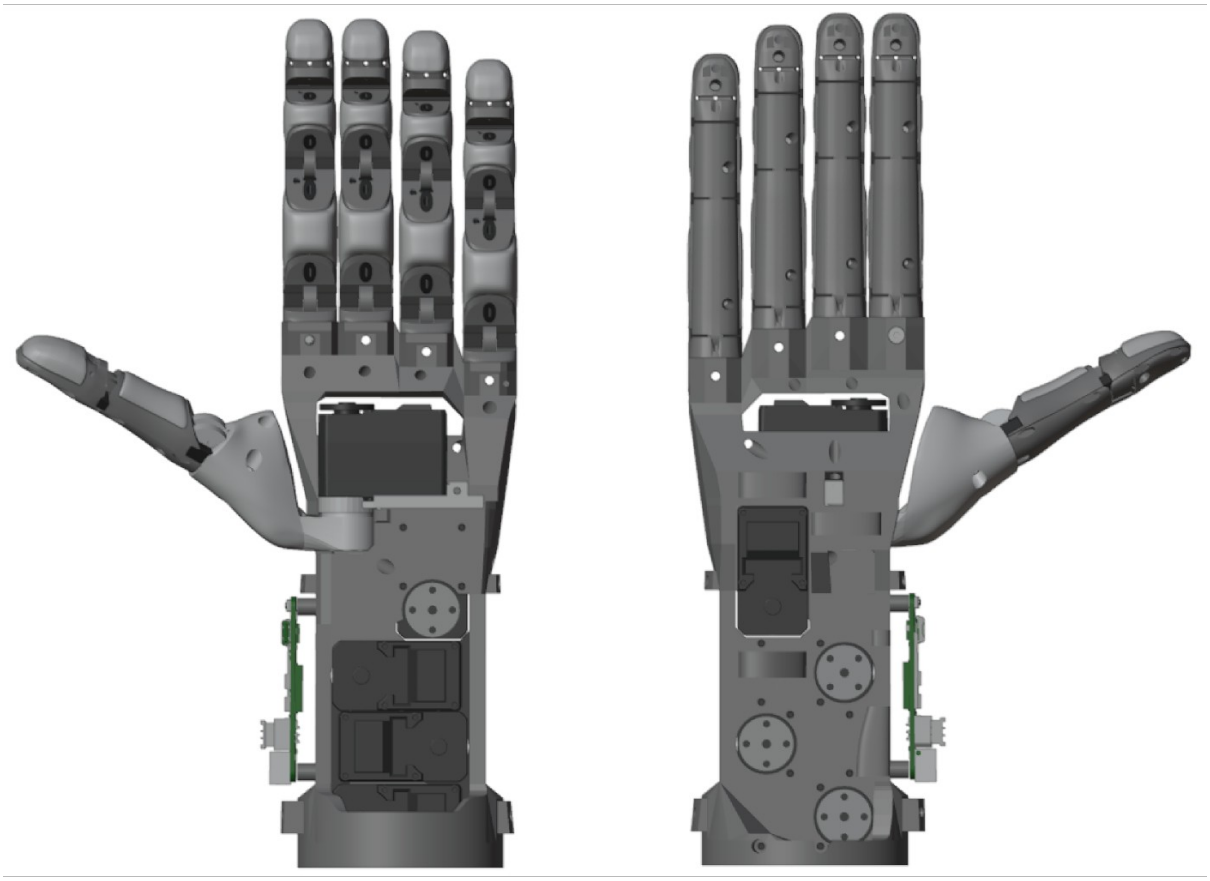


Figure 5.22: Fingers installed, front/rear view

5.7 Installing the Bowden tubes

5.7.1 Installing the Bowden tube for the Index finger

For the INDEX finger only:

1. Flexion tendon:
    1. Insert the PTFE tube into the outlet hole (on the rear of the skeleton) of the flexion tendon (figure 5.23), let it come out through the inlet hole on the front of the skeleton and insert it into the initial slot (figure 5.24)
    2. Take the remaining part (on the rear of the skeleton) and insert it into the final slot shown in figure 5.25.
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
    1. Insert one end of the PTFE tube into the initial slot (on the rear of the skeleton) of the extension tendon (figure 5.27)
    2. Insert the other end into the final slot shown in figure 5.26: if the tube is too long, shorten it by cutting it with a utility knife.

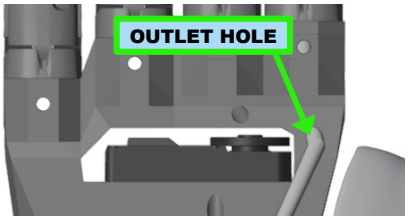


Figure 5.23: Inserting the tube into the tendon outlet hole (rear view)

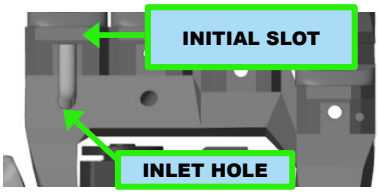


Figure 5.24: Inserting the tube into the initial slot (front view)

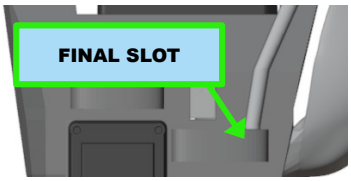


Figure 5.25: Inserting the tube into the final slot (rear view)

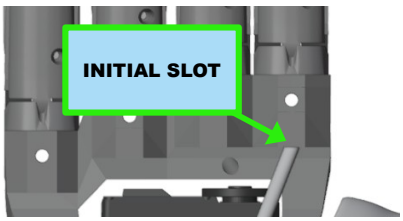


Figure 5.27: Inserting the tube into the initial slot (rear view)

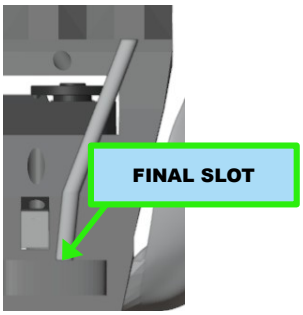


Figure 5.26: Inserting the tube into the final slot (rear view)

5.7.2 **Installing the Bowden tube for the Middle finger**

For the MIDDLE finger only:

1. Flexion tendon:
    1. Insert the PTFE tube into the outlet hole (on the rear of the skeleton) of the flexion tendon (figure 5.30), let it come out through the inlet hole on the front of the skeleton and insert it into the initial slot (figure 5.28)
    2. Take the remaining part (on the rear of the skeleton) and thread it first through the 2 pass-through tunnels and then into the final slot (figure 5.29).
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
    1. Insert one end of the PTFE tube into the initial slot (on the rear of the skeleton) of the extension tendon (figure 5.32)

Insert the other end into the final slot shown in figure 5.31

2. Note: if the tube is too long, shorten it by cutting it with a utility knife.

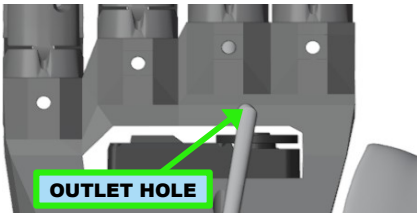


Figure 5.30: Inserting the tube into the tendon outlet hole (rear view)

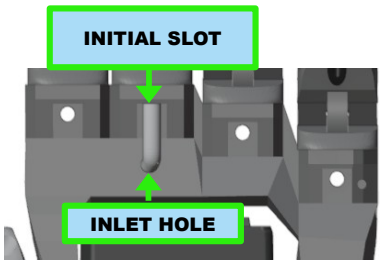


Figure 5.28: Inserting the tube into the circular slot (front view)

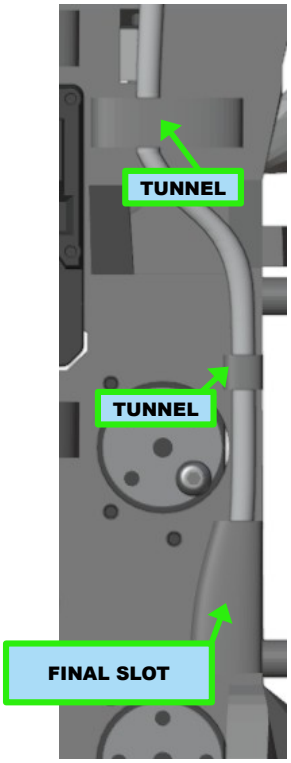


Figure 5.29 : Inserting the tube into the two pass-through tunnels and then into the final slot (rear view)

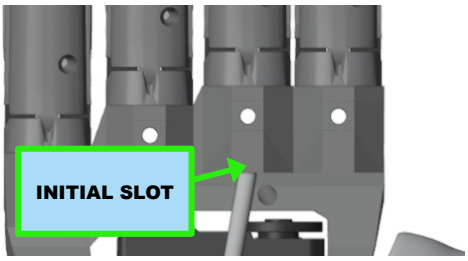


Figure 5.32 : Inserting the tube into the initial slot (rear view)

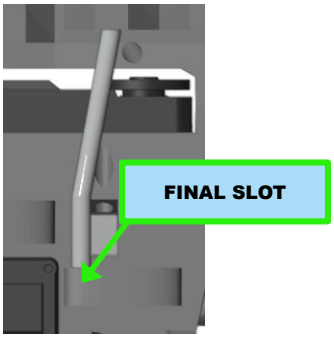


Figure 5.31 : Inserting the tube into the final slot (rear view)

5.7.3 **Installing the Bowden tube for the Ring finger**

For the RING finger only

1. Flexion tendon:
    1. Insert the PTFE tube into the outlet hole (on the rear of the skeleton) of the flexion tendon (figure 5.33), let it come out through the inlet hole on the front of the skeleton and insert it into the initial slot (figure 5.35)
    2. Take the remaining part (on the rear of the skeleton) and thread it first through the pass-through tunnel and then into the final slot (figure 5.34).
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
    1. Insert one end of the PTFE tube into the initial slot (on the rear of the skeleton) of the extension tendon (figure 5.37)
    2. Insert the other end into the pass-through tunnel and then into the final slot (figure 5.36).
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.

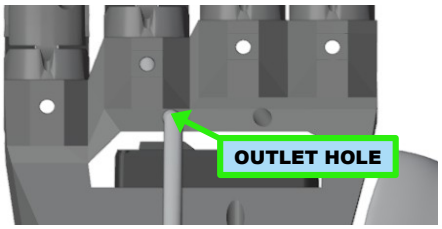


Figure 5.33: Inserting the tube into the tendon outlet hole (rear view)

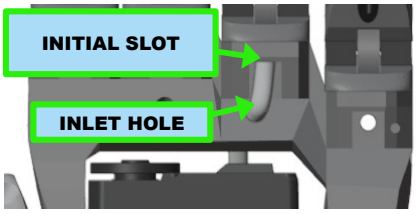


Figure 5.35 : Inserting the tube into the initial slot (front view)



Figure 5.34: Inserting the tube into the pass-through tunnel and then into the final slot (rear view) (vista posteriore)

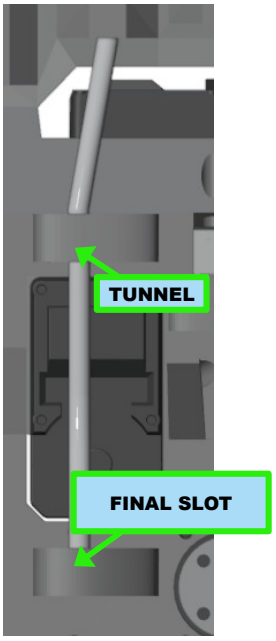


Figure 5.36: Inserting the tube into the pass-through tunnel and then into the final slot (rear view)

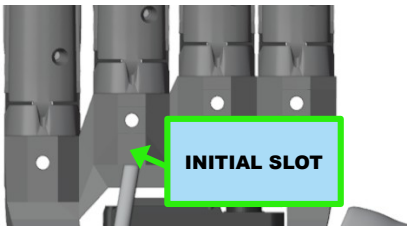


Figure 5.37: Inserting the tube into the initial slot (rear view)

5.7.4 **Installing the Bowden tube for the Little finger**

For the LITTLE finger only

1. Flexion tendon:
    1. Insert the PTFE tube into the outlet hole (on the rear of the skeleton) of the flexion tendon (figure 5.40), pass it through the pass-through tunnel and let it come out through the inlet hole on the front of the skeleton and insert it into the initial slot (figure 5.39).
    2. Take the remaining part (on the rear of the skeleton) and thread it first through the pass-through tunnel and then into the final slot (figure 5.38).
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
    1. Insert one end of the PTFE tube into the initial slot (on the rear of the skeleton) of the extension tendon (figure 5.42).
    2. Insert the other end into the pass-through tunnel and then into the final slot (figure 5.41).
    3. Note: if the tube is too long, shorten it by cutting it with a utility knife.

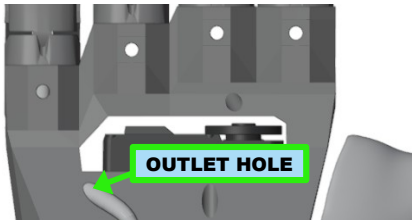


Figure 5.40 : Inserting the tube into the tendon outlet hole (rear view)

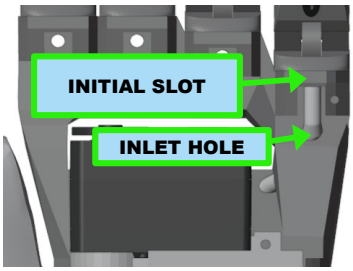


Figure 5.39 : Inserting the tube into the initial slot (front view)



Figure 5.38 : Inserting the tube into the pass-through tunnel and then into the final slot (rear view) (vista posteriore)

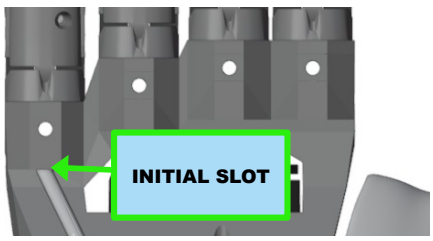


Figure 5.42 : Inserting the tube into the initial slot (rear view)

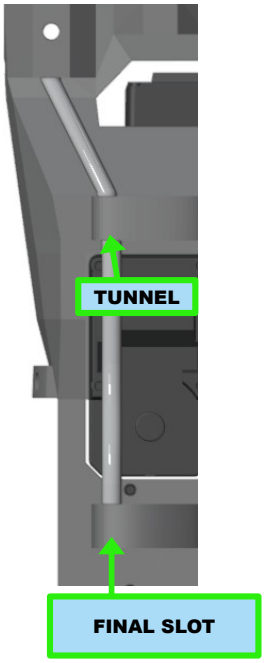


Figure 5.41 : Inserting the tube into the pass-through tunnel and then into the final slot (rear view)

5.7.5 **Checking the Bowden tube installation**

Before proceeding, make sure all the Bowden tubes are correctly installed as shown in the figure.

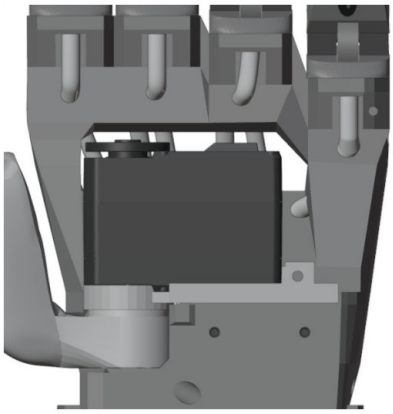


Figure 5.44: Front view with Bowden tubes installed

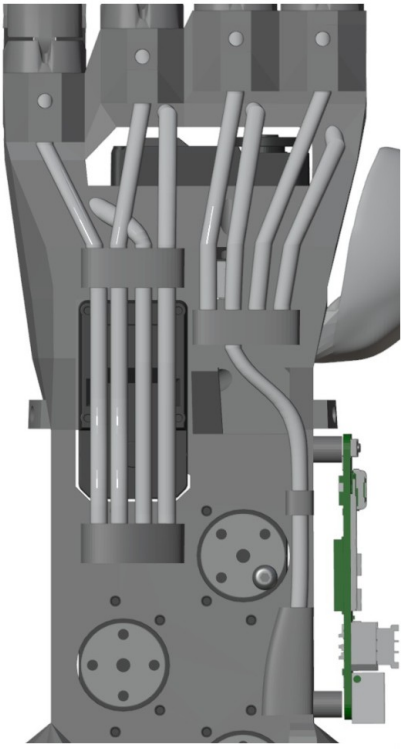


Figure 5.43: Rear view with Bowden tubes installed

5.8 Inserting tendons into the Bowden tubes (flexion end)

Procedure to be applied only to the tendons of the index, middle, ring, and little fingers.

This operation requires a nylon needle at least 20 cm long (40 cm of unfolded thread).

For each finger:

1. On the front of the skeleton, insert the two ends of the nylon needle into the inlet hole in front of the finger slot (figure 5.45) and push them into the PTFE tube below, bringing them out through the outlet hole on the rear of the skeleton (figure 5.46)
2. Insert the flexion end of the tendon into the loop of the nylon needle left outside the inlet hole (figure 5.45) and pull the ends of the nylon needle from the other end of the Bowden tube until the entire tendon comes out of the outlet hole (figure 5.47).

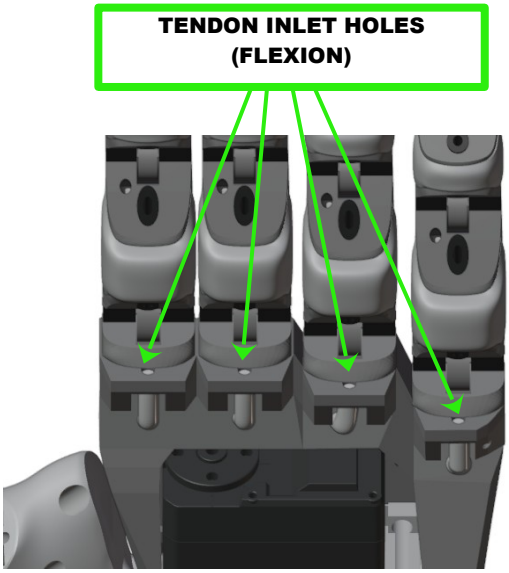


Figure 5.45: Tendon inlet holes (flexion)

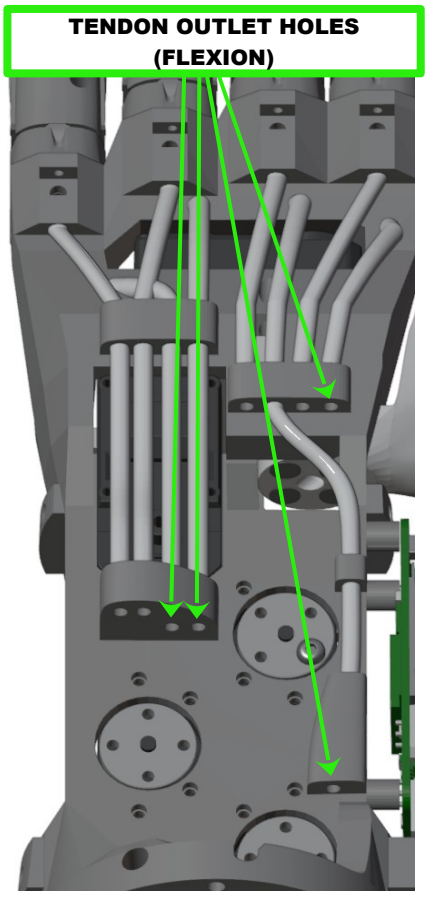


Figure 5.46: Tendon outlet holes (flexion)

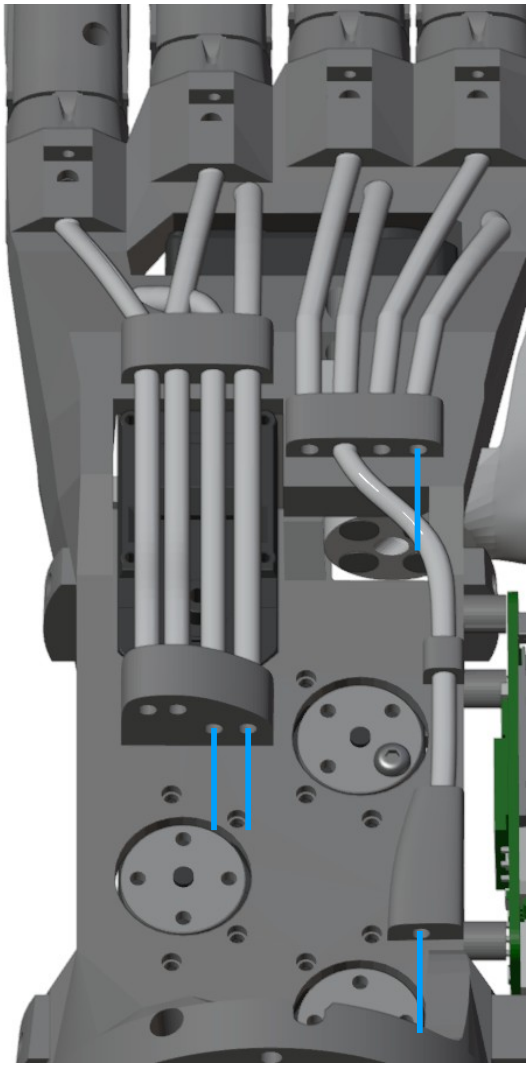


Figure 5.47: Flexion ends of the tendons emerging from the Bowden tubes

5.9 Inserting tendons into the Bowden tubes (extension end)

Procedure to be applied only to the tendons of the index, middle, ring, and little fingers.

This operation requires a nylon needle at least 20 cm long (40 cm of unfolded thread).

For each finger:

3. On the rear of the skeleton: insert the two ends of the nylon needle into the finger's final hole and then into the inlet hole of the Bowden tube below (figure 5.49).
4. Push the nylon threads until they come out of the outlet hole (figure 5.48).
5. Insert the extension end of the tendon into the loop of the nylon needle left outside the finger's final hole (figure 5.49) and pull the ends of the nylon needle from the other end of the Bowden tube until the entire tendon comes out of the outlet hole (figure 5.50)

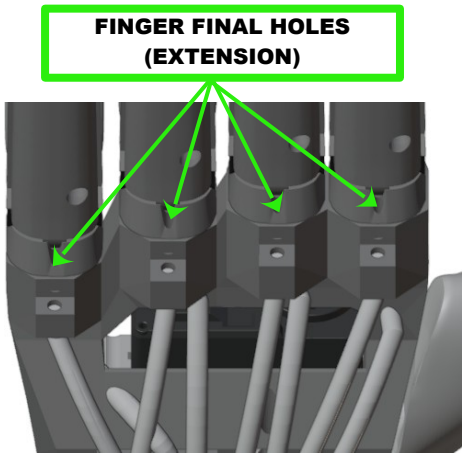


Figure 5.49: Tendon inlet holes (extension)

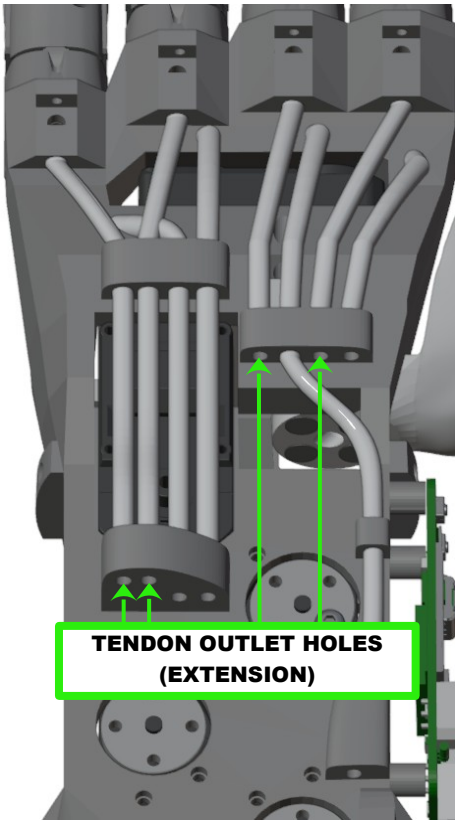


Figure 5.48: Tendon outlet holes (extension)

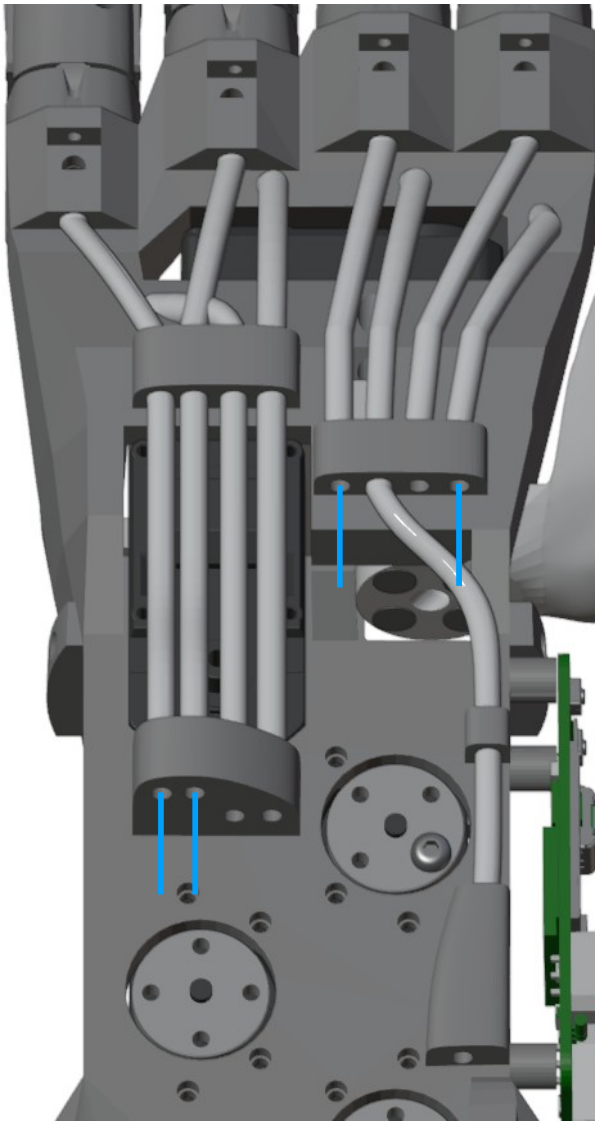


Figure 5.50: Extension ends of the tendons emerging from the Bowden tubes

5.10 Installing tendons on the spools

5.10.1 Safety information

!! DO NOT PULL THE TENDON WITH BARE HANDS !!

MANDATORY

- Use a vice to hold the part being worked on firmly in place
- Use assembly pliers to pull the tendon

POSSIBLE HARM:

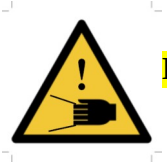
- **Severe cuts to the fingers**
- **Blood loss**



MANDATORY USE OF THE VICE



MANDATORY USE OF THE ASSEMBLY PLIERS



DANGER OF FINGER CUTS

5.10.2 Nylon needle technique

The tendons are thin and difficult to thread, which is why a special needle must be used.

This special needle is actually a length of nylon thread used as if it were a needle.

This method was chosen after multiple trials with various different tools, as it proved to be the fastest, most reliable, and safest.

The procedure for threading the tendon through a hole using a nylon thread (10 cm long, 0.35 mm diameter) as a needle is shown below:



WARNING: the tendons can cut fingers!

MANDATORY: use the vice to secure the spool, and the assembly pliers to pull the tendon.

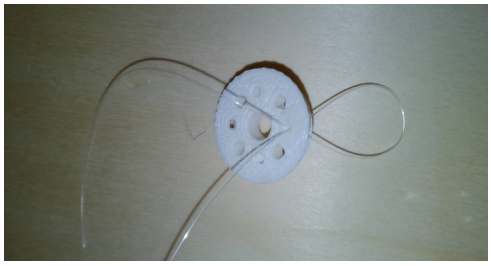


Figure 5.51: Threading the ends of the nylon thread, one at a time, through the hole

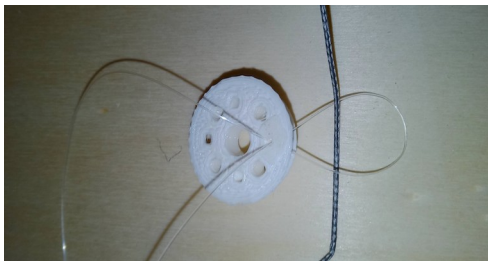


Figure 5.52: Threading the tendon into the nylon loop



Figure 5.53: Pulling with the assembly pliers the two ends of the nylon loop through the other side of the hole

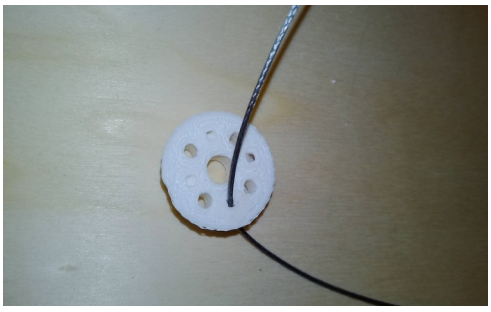


Figure 5.54: pulling with the assembly pliers the shorter part of the tendon through the other side of the hole

NOTE: Hereafter, this sequence of operations will simply be referred to as "threading the tendon." It is implicitly assumed that this is done using the nylon needle.

5.10.3 Installing tendons on the spools (index, middle, ring and little fingers)



WARNING: the tendons can cut fingers!

MANDATORY: use the vice to secure the skeleton, and the assembly pliers to pull the tendon.

5.10.3.1 Preparing the spool assembly

Each servo requires one flexion spool and one extension spool, mounted one above the other. See figure.

For each flexion spool (shown in yellow in figure 5.56):

1. Position the extension spool (shown in blue in figure 5.56)
    1. Make sure the flexion spool (yellow in figure 5.52) has its smaller diameter at the bottom.
2. Fasten the extension spool with 4 M3 button-head screws, 10 mm long (figure 5.55): the screws will temporarily hold the assembly together by a simple snap fit, without tightening (it is important that they remain joined until the end of the next phase 'Installing the tendons on the spools').



Figure 5.56: Assembled spool



Figure 5.55: The bolts hold the assembly together

5.10.3.2 Installing tendons on the spools (index, middle, ring and little fingers)

Make sure each spool assembly is correctly assembled (the screws temporarily hold the extension spool joined to the flexion spool by simple interlocking, without being tightened)

Preparation:

1. Make sure all servo motors are connected, and that the spools are not installed.
2. Power the logic board using the power connector (make sure the USB cable is disconnected) as shown in figure 5.57.
3. Connect (BT SERIAL TOOL TO BE PROVIDED) via Bluetooth to the device (device name: YEAH-HAND-X-YYYYYY, where X is 'L' or 'R' depending on the model, and YYYYYY is the serial number)
4. send the 'INSTALL' selection serial command via Bluetooth.
5. wait for the servo motors to complete their movement, then proceed with the following paragraphs:
   1. ***Installing tendons on the spools (index, middle, ring and little fingers)***
   2. ***Installing tendons on the spools (thumb)***

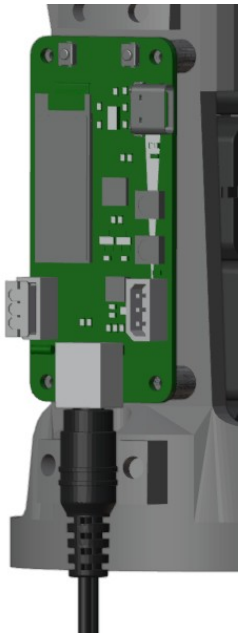


Figure 5.57: Power connector inserted

Installing tendons on the spools (index, middle, ring and little fingers):

1. For each servo motor:
    1. Identify the corresponding finger (or fingers, for ring and little) and the corresponding spool assembly (the flexion spool and extension spool are currently held together by inserted, but not yet tightened, bolts)
    2. Take the flexion end of the tendon (2 ends for ring and little)
    3. Thread the end(s) into the bottom-right hole of the flexion spool and pass it out through the corresponding hole of the extension spool (figure 5.59), make sure enough tendon protrudes to prevent it slipping back through.
    4. Make sure the spool assembly does not come apart during these operations.
    5. Take the extension end of the tendon (2 ends for ring and little)
    6. Thread the end(s) into the top hole of the extension spool and pass it out through the bottom hole of the same spool (figure 5.58), make sure enough tendon protrudes to prevent it slipping back through.
    7. Position the spool assembly onto its servo motor, with the arrow pointing up (figure 5.60)
    8. Screw the spool assembly onto the flange of its servo motor using the M3 bolts
    9. Completing the extension end of the tendon:
        1. Make the tendon wrap one clockwise turn around the extension spool before it enters the top hole (figure 5.61)
        2. Pull the tendon through the bottom hole (until the finger is straight) and wrap it around the left bolt with a clockwise rotation (figure 5.62).
        3. While holding the tendon taut with the assembly pliers, firmly tighten the bolt with the wrench to lock the tendon (figure 5.63)
    10. Completing the flexion end of the tendon:
        1. Pull the tendon until taut, without shifting the position of the controlled finger, and wrap it around the right bolt with a clockwise rotation (figure 5.64).
        2. While holding the tendon taut with the assembly pliers, firmly tighten the bolt to lock the tendon (figure 5.65).

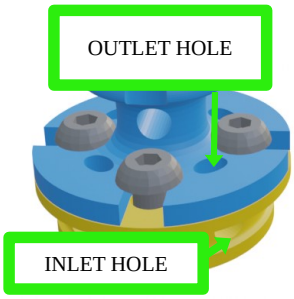


Figure 5.59: Inserting the flexion end

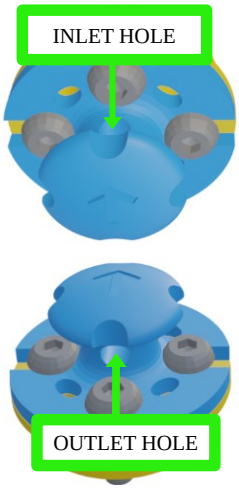


Figure 5.58: Inserting the extension end

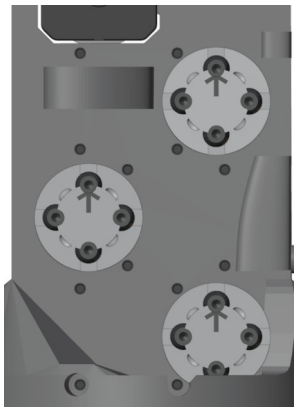


Figure 5.60: Spool assemblies mounted with arrow pointing up

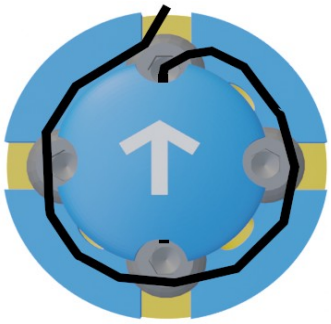


Figure 5.61: Pre-winding operation for the tendon's extension end

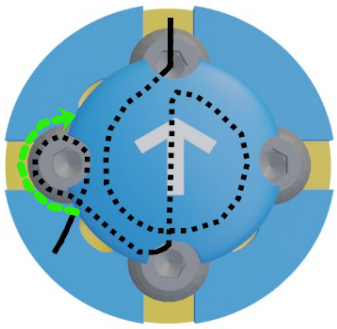


Figure 5.62: Extension end: Rotation around the bolt

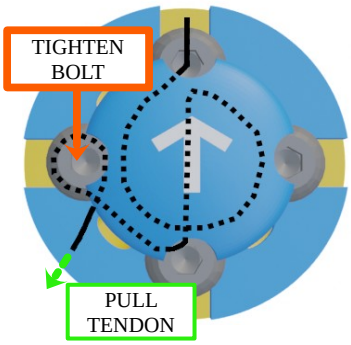


Figure 5.63: Extension end – Pulling and securing

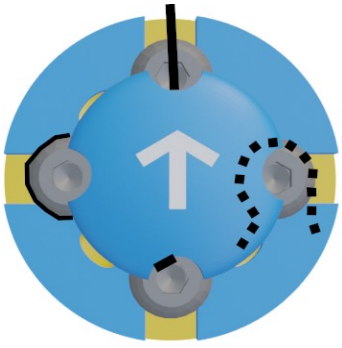


Figure 5.64: Flexion end: Rotation around the bolt

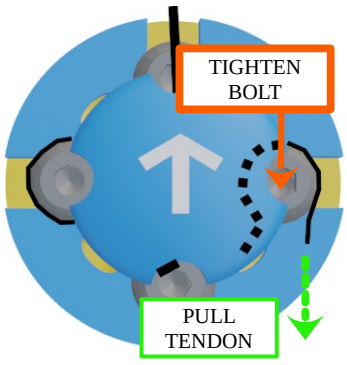


Figure 5.65: Flexion end: pulling and securing

Installing tendons on the spools (thumb)

Note: the thumb uses a single spool for both extension and flexion.

1. Only for the thumb servo motor
    1. Take the flexion end of the tendon
    2. Thread the end into the top right hole of the spool and pass it out through the corresponding outlet hole (figure 5.66), make sure enough tendon protrudes to prevent it slipping back through.
    3. Take the extension end of the tendon
    4. Thread the end into the bottom left hole of the spool and pass it out through the corresponding outlet hole of the same spool (figure 5.67), make sure enough tendon protrudes to prevent it slipping back through.
    5. Position the spool on its servo motor and fasten it with 4 M3 button-head screws with hex socket, 10 mm long (figure 5.68)
    6. Pull the extension end of the tendon through the bottom hole (until the finger is straight) and secure it to the bottom bolt with a counter-clockwise rotation around it (figure 5.69). While holding the tendon taut with the assembly pliers, firmly tighten the bolt to lock the tendon (figure 5.69). Figure 5.70 shows the completed operation.
    7. Pull the flexion end of the flexion tendon without shifting the finger's position and secure it to the bolts as shown in figure 5.72. While holding the tendon taut with the assembly pliers, firmly tighten the bolt to lock the tendon (figure 5.72). Figure 5.71 shows the completed operation.

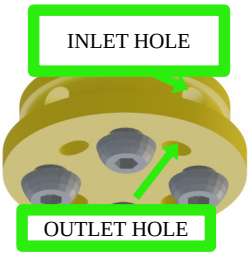


Figure 5.66: Thumb spool: Inserting the flexion end

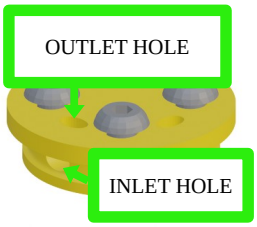


Figure 5.67: Thumb spool: Inserting the extension end

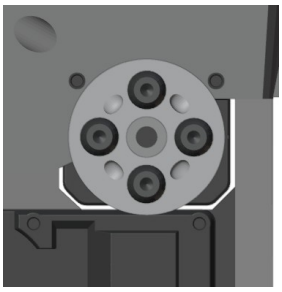


Figure 5.68: Positioning the spool on the servo motor

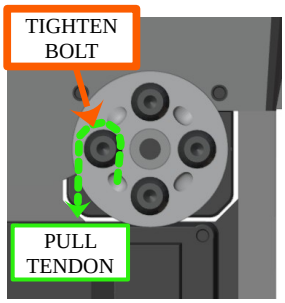


Figure 5.69: Rotation around the bolt and securing

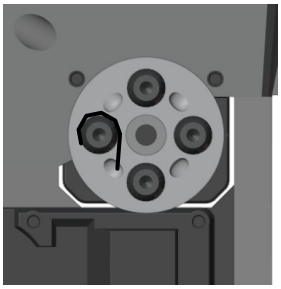


Figure 5.70: Tendon secured under the bolt

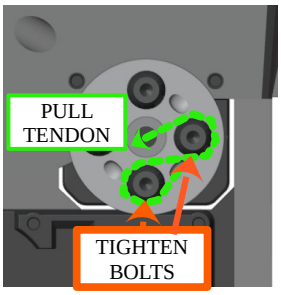


Figure 5.72: Rotation around the bolt

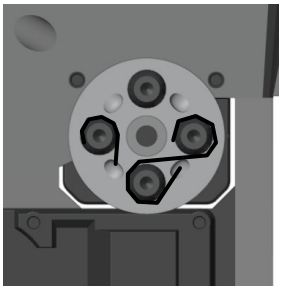


Figure 5.71: Tendon secured under the bolt

5.11 **Installing the palm**

Position the palm and fasten it with 4 M2 self-tapping screws, 6.5mm long (see figure)

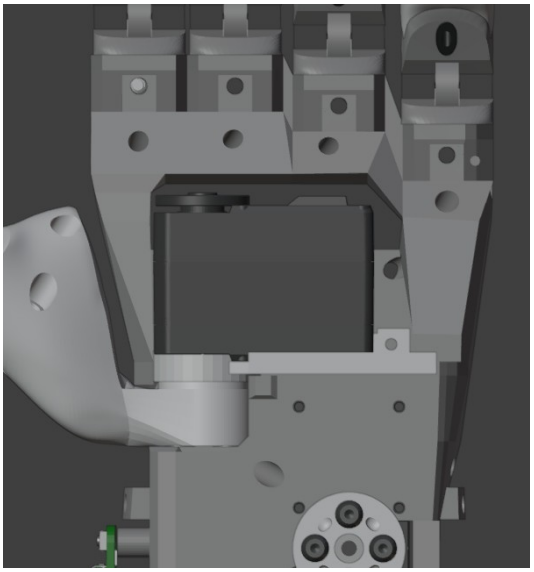


Figure 5.73: Skeleton - palm area

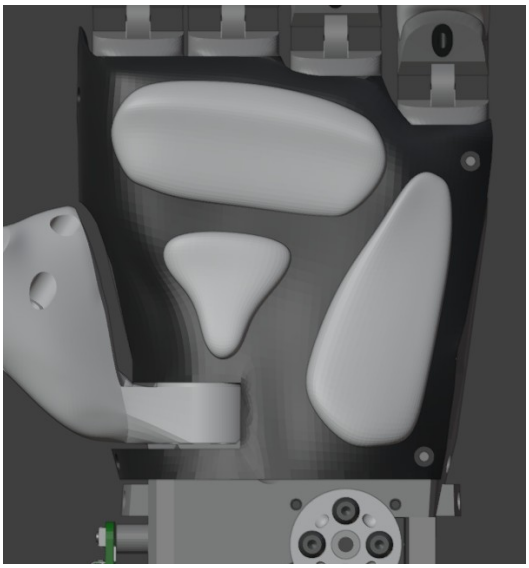


Figure 5.74: Skeleton - palm installed

5.12 Installing the covers

5.12.1 Installing the upper cover

Position the pins on the thumb side into the skeleton holes, snap them into place, then, while rotating the cover, snap the pins on the other side into place (see figures 5.76 and 5.75).

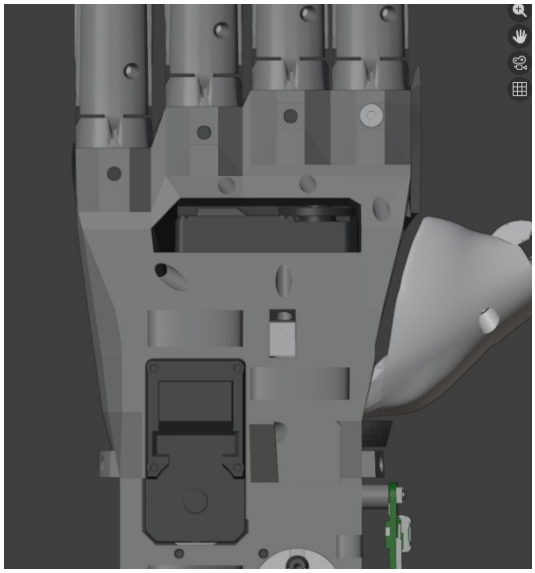


Figure 5.76: Skeleton - Upper cover area

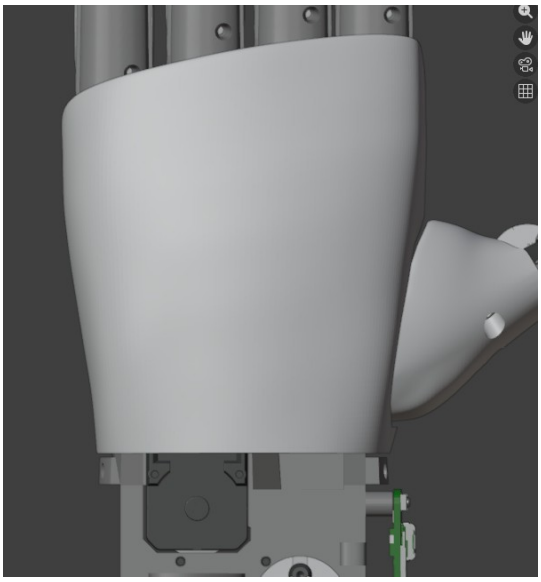


Figure 5.75: Skeleton - Upper cover installed

5.12.2 Installing the lower covers

1. Make sure the logic board is already mounted on the skeleton
2. Make sure the logic board is connected to the servo motor at the base of the skeleton with the control cable
3. Position the pins on one side of the rear cover onto the corresponding interlocking holes on the skeleton, snap them into place, then, while rotating the cover, snap the pins on the other side into place. See figures 5.77 and 5.78.
4. Position the pins on one side of the front cover onto the corresponding interlocking holes on the skeleton, snap them into place, then, while rotating the cover, snap the pins on the other side into place. See figures 5.79 and 5.80.

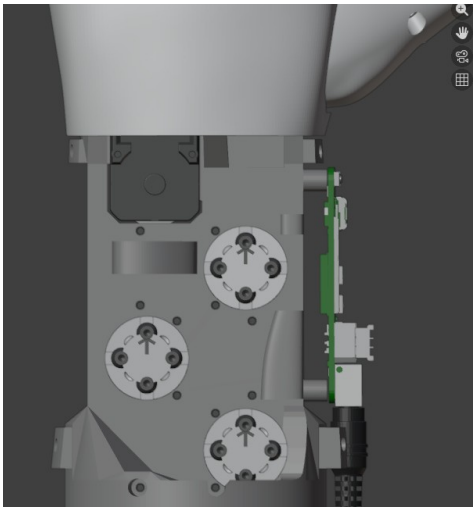


Figure 5.77: Skeleton - Rear lower cover area

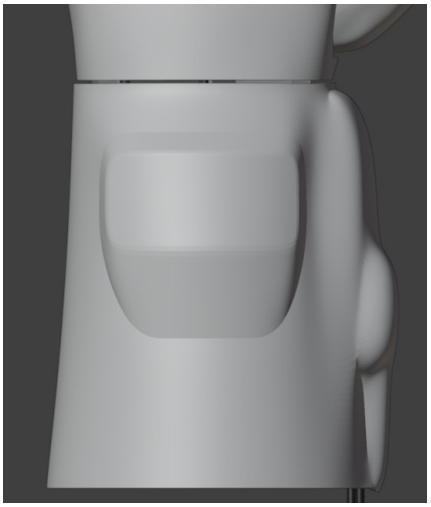


Figure 5.78: Skeleton - Rear lower cover installed

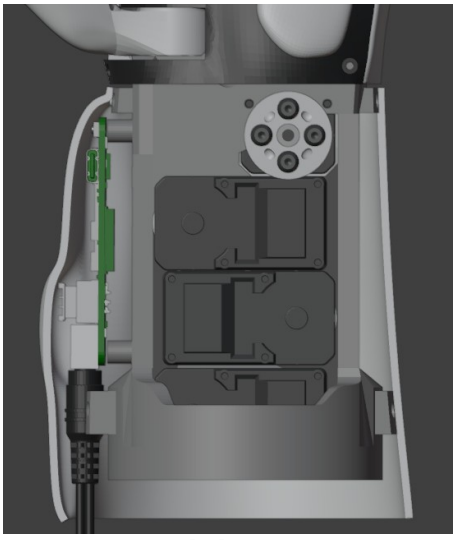


Figure 5.79: Skeleton - Front lower cover area



Figure 5.80: Skeleton - Front lower cover installed

5.12.3 Installing the finger covers

For each upper finger (index, middle, ring, little)

1. position the distal cover on the distal phalanx and press to snap it into place
2. position the medial cover on the medial phalanx and press to snap it into place
3. position the proximal cover on the proximal phalanx and press to snap it into place

For the thumb:

1. position the distal cover on the distal phalanx and press to snap it into place
2. position the proximal cover on the proximal phalanx and press to snap it into place

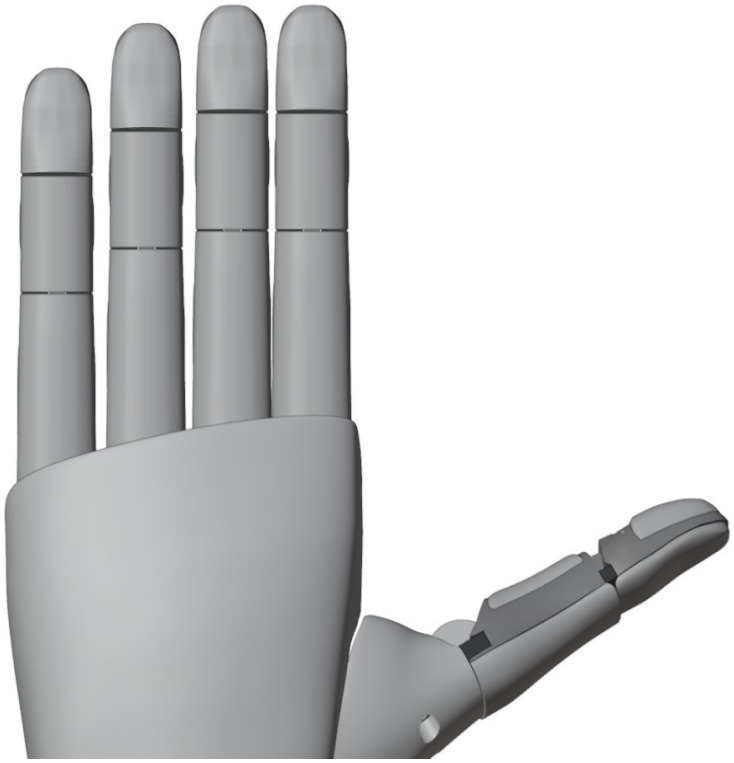


Figure 5.81: Finger covers installed

6 Installing the robotic hand on a Robotic Manipulator

1. Position the flange on the robot wrist, aligning it with the alignment pin.
2. Tighten with 4 M6 countersunk bolts, 3 cm long. TBD
3. Position the robotic hand on the flange in the desired orientation.
4. Secure with 4 M3 countersunk bolts, 3 cm long. TBD
5. Connect the robotic hand's power cable connector to the robot wrist connector.

The supplied logic board (WaveShare ESP32 Servo Driver Board) requires 12V DC.
If the robotic hand needs to be powered at 24V DC, the voltage conversion board must be installed, which steps the voltage down from 24V to 12V. In that case, a 2A fuse must be used.



USE ONLY POWER SUPPLIES WITH THE FOLLOWING OUTPUT CHARACTERISTICS:
VOLTAGE: 12V DC
2A < CURRENT < 5A

Depending on the available power supply, install the appropriate fuse (see table).

DC power supply	Fuse to install
12V 2A	2A
12V 4A	4A
12V 5A	5A
24V 2A + 12V Conversion Board 4A	4A
24V 1A + 12V Conversion Board 2A	2A

TODO INSERT FIGURES OF THE 24/12 CONVERSION BOARD

7 Original equipment manufacturer settings

The robotic hand's default settings are as follows:

- Control mode
  - Grasp Control
    - MONKEY GRASP
- Finger position:
  - factor: 0% (maximum opening)
- Bluetooth device name:
  - YeahHand_XXX, where XXX is a unique code.

8 Operation



READ THE MANUAL BEFORE USING THE ROBOTIC HAND



READ THE INFORMATION ON TYPICAL OPERATION

8.1 Startup

To start the robotic hand, activate power from the controls of the robotic arm on which it is installed, so that the robot wrist connector supplies 12V and at least 162W of power.

The robotic hand begins the tendon auto-calibration phase; once complete, the robotic hand will be in the maximum finger-opening position, in grasping mode: monkey grasp.

Connect to the robotic hand from the robot's control system via a Bluetooth serial character connection (device name: YEAH-HAND-L_XXXXX or YEAH-HAND-R_XXXXX, where XXXXX is the robotic hand's serial number).

Serial commands follow the format CMD X Y ...\r\n, where CMD is the command string and X Y ... are additional parameters, while \r\n is the final send sequence (carriage return and newline)

Configuration

The following parameters can currently be configured:

Parameter	Command	Variables
PINCH Offset	SETPINCH X\r\n	X: [-40, 40] Pinch Offset
Load Limit	LIM X Y\r\n	X: [1, 5] Motor ID Y: [0, 600] Load Limit Value

8.2 Selecting the control mode

Control mode	Full command
Monkey grasp	MONKEY\r\n
Power grasp	POWER\r\n
Pinch grasp [1]	PINCH\r\n
Relax	RELAX\r\n
RAW	RAW\r\n

[1] The PINCH gesture requires configuring the pinch offset via the SETPINCH command, see the relevant paragraph.

8.3 Finger movement

8.3.1 Grasp Control mode

If a mode among MONKEY, POWER, POWERSMALL, PINCH has been selected, the robotic hand can be commanded as follows:

- Command: X\r\n
- Meaning: X expresses the integer value [0, 100] representing the percentage of finger closure (each type of selected grasp applies closure to all fingers differently from the other grasps)

Examples:

EXAMPLE	FULL COMMAND
Maximum opening	0\r\n
Maximum closure	100\r\n
50% closure	50\r\n
75% closure	75\r\n

8.3.2 **ROS mode**

If the 'ROS' mode has been selected, the robotic hand can be commanded as follows:

- Command: ROS V W X Y Z\r\n
- Meaning:
  - ROS : initial data packet tag
  - V expresses an integer closure value [0, 100] for index flexion
  - W expresses an integer closure value [0, 100] for middle flexion
  - X expresses an integer closure value [0, 100] for ring flexion
  - Y expresses an integer closure value [0, 100] for thumb flexion
  - Z expresses an integer closure value [0, 100] for thumb rotation

Examples:

EXAMPLE	**FULL COMMAND**
Index flexion: 30% Middle flexion: 50% Ring and little flexion: 10% Thumb flexion: 70% Thumb rotation: 65%	ROS 30 50 10 70 65\r\n
Index flexion: 10% Middle flexion: 20% Ring and little flexion: 30% Thumb flexion: 40% Thumb rotation: 50%	ROS 10 20 30 40 50\r\n
Flexion: maximum opening Thumb rotation: maximum opening (open hand)	ROS 0 0 0 0 0\r\n
Flexion: maximum closure Thumb rotation: maximum closure (thumb in opposition)	ROS 100 100 100 100 100\r\n

9 Changing product or capacity

Before replacing an object gripped by the robotic hand, make sure its characteristics fall within the limits set out in the chapter Technical characteristics of the machine in the section "Object properties".

10 Inspections, tests and maintenance

10.1 Expected component lifetime

Note: a cycle is one complete closure of the finger from the 0% position (fully open) to the 100% position (fully closed, equal to 300° of servo motor rotation) and subsequent reopening to the 0% position.

COMPONENT	LIFETIME IN CYCLES[2]
Skeleton	200000
TRS Platform	200000
Thumb base	20000
Fingers	20000
Spools	20000
Covers	200000
Tendons and Bowden tubes	200000
Servo motors [3]	20000
Logic board	200000

10.2 Nature, frequency and inspection criteria

COMPONENT	INSPECTION INTERVAL IN CYCLES
Skeleton	50000
TRS Platform	50000
Thumb base	5000
Fingers	5000
Spools	5000
Covers	50000
Tendons and Bowden tubes	50000
Servo motors	5000
Logic board	50000

[2] Values found experimentally with prolonged use of the product (2024-2025-2026 period).
[3] Calculated based on data from the original manufacturer WaveShare. The ST3215-HS servo motor has a service life of 100000 cycles, where one cycle consists of rotating the flange from 0° to 60° and back to 0°.

10.3 **Spare parts specifications**

For photos and images of the components, see chapter 14.1 Components

All spare parts are specified in the following table:

DESCRIPTION	QTY PER PRODUCT	MANUFACTURER	MODEL
Left hand skeleton	1	VITTORIO-LUMARE	SKELETON-L
Right hand skeleton	1	VITTORIO-LUMARE	SKELETON-R
Index finger with silicone pads	1	VITTORIO-LUMARE	INDEX
Middle finger with silicone pads	1	VITTORIO-LUMARE	MIDDLE
Ring finger with silicone pads	1	VITTORIO-LUMARE	RING
Little finger with silicone pads	1	VITTORIO-LUMARE	LITTLE
Thumb with silicone pads	1	VITTORIO-LUMARE	THUMB
Left hand palm	1	VITTORIO-LUMARE	PALM-L
Right hand palm	1	VITTORIO-LUMARE	PALM-R
Left hand thumb base	1	VITTORIO-LUMARE	THUMB-BASE-L
Right hand thumb base	1	VITTORIO-LUMARE	THUMB-BASE-R
Bowden tubes	1	VITTORIO-LUMARE	TENDON-TUBING
Index finger flexion tendon spool	3	VITTORIO-LUMARE	FLEXION-SPOOL
Index finger extension tendon spool	3	VITTORIO-LUMARE	EXTENSION-SPOOL
Thumb tendon spool	1	VITTORIO-LUMARE	THUMB-SPOOL
Left hand upper cover	1	VITTORIO-LUMARE	UPPER-COVER-L
Right hand upper cover	1	VITTORIO-LUMARE	UPPER-COVER-R
Left hand lower front cover	1	VITTORIO-LUMARE	LOWER-COVER-FRONT-L
Left hand lower rear cover	1	VITTORIO-LUMARE	LOWER-COVER-BACK-L
Right hand lower front cover	1	VITTORIO-LUMARE	LOWER-COVER-FRONT-R
Right hand lower rear cover	1	VITTORIO-LUMARE	LOWER-COVER-BACK-R
Index finger cover set	1	VITTORIO-LUMARE	INDEX-COVERS
Thumb cover set	1	VITTORIO-LUMARE	THUMB-COVERS
Flange ISO-9409-1-50-4-M6	1	VITTORIO-LUMARE	FLANGE-ISO-9409-1-50-4-M6
Flange ISO-9409-1-40-4-M6	1	VITTORIO-LUMARE	FLANGE-ISO-9409-1-50-4-M6
Flange PCD-40-4-M5	1	VITTORIO-LUMARE	FLANGE-PCD-40-4-M5
Flange PCD-40-4-M6	1	VITTORIO-LUMARE	FLANGE-PCD-40-4-M6
Flange Agilex Black	1	VITTORIO-LUMARE	FLANGE-AGILEX-NERO
Servo motors and cables	5	WAVESHARE	ST3215-HS
Logic board	1	WAVESHARE	Servo Driver with ESP32
Polyethylene tendons (skein)	1	HERCULES	90.7 kg, 0.75mm, 8 strands
M3 screws, countersunk head, hex socket, 20 mm long	**TBD**	**N.A.**	**N.A.**
M3 screws, countersunk head, hex socket, 16 mm long	**TBD**	**N.A.**	**N.A.**
M3 screws, countersunk head, hex socket, 12 mm long	**TBD**	**N.A.**	**N.A.**
M3 screws, button head, hex socket, 10 mm long TODO CHECK	**TBD**	**N.A.**	**N.A.**
M6 screws, countersunk head, hex socket, 35 mm long	**TBD**	**N.A.**	**N.A.**
M2 screws, cylindrical head with hex socket, 5 mm long	**TBD**	**N.A.**	**N.A.**

M2 self-tapping screws, Phillips button head, 6.5mm long	**TBD**	**N.A.**	**N.A.**

10.4 Tools and equipment

10.4.1 Assembly and maintenance tools and equipment

These are the tools and equipment required for assembly

DESCRIPTION	QTY	MANUFACTURER	MODEL
Long Phillips #1 screwdriver for electronics 6.69"	1	WIHA	261110
Precision pliers 4.72"	1	ADAFRUIT INDUSTRIES LLC	422
Hex screwdriver 2mm 5.71"	1	WIHA	26320
Hex screwdriver 1.5mm 7.05"	1	WIHA	36715
Small assembly pliers	1	OLIMEX LTD	CHN-PLI-LC

10.5 Energy isolation procedures

To disconnect the electrical power source feeding the robotic hand, it is sufficient to unplug the power plug from the side connector (figure 10.1).

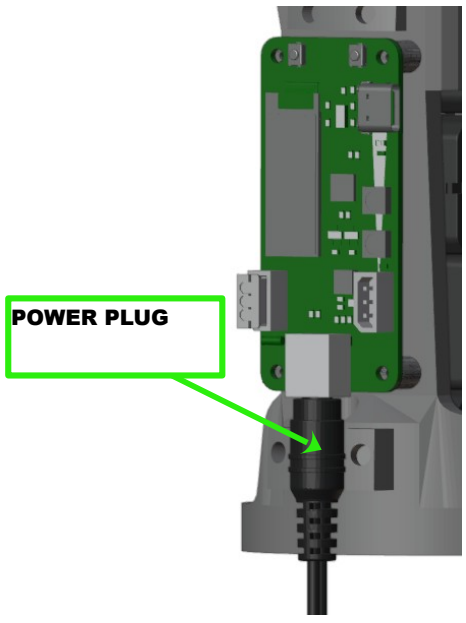


Figure 10.1: Connector with power plug inserted

10.6 **Extraordinary maintenance**

These operations are to be performed only when a component requires replacement.

10.6.1 **Finger replacement**

Perform in sequence:

1. Tendon removal
2. Finger removal
3. Finger installation
4. Tendon installation
5. Tendon registration

10.6.2 **Finger removal**


USE THE VICE

1. Unscrew all finger fastening bolts
2. Remove all fingers from their slots

10.6.3 **Finger installation**


USE THE VICE

See Chapter 5.5. Installing the Index, Middle, Ring and Little fingers

See Chapter 5.1.5. Installing the Thumb finger

10.6.4 Tendon removal



1. Cut all tendons with scissors
2. Unscrew the tendon fastening bolts from all spools using the 2.5mm hex wrench provided
3. Remove all tendons using the assembly pliers provided

10.6.5 Tendon installation



See Chapter 5.4. Installing the tendons in the fingers

10.6.6 Motor replacement

Perform in sequence:

1. Motor removal
2. Motor installation

10.6.7 Motor removal

1. Remove each servo motor by unscrewing its 4 fastening screws, following this sequence:
    1. ring-little finger flexion servo motor
    2. index finger flexion servo motor
    3. middle finger flexion servo motor
    4. thumb flexion servo motor
    5. thumb rotation servo motor

10.6.8 Motor installation

See Chapter 5.1. Servo motor installation and registration

10.6.9 Cover replacement

Perform in sequence:

1. Cover removal
2. Cover installation

10.6.10 Cover removal

1. Detach the front snap-fit cover (palm side)
2. Detach the rear snap-fit cover (back side)

10.6.11 Cover installation

See Chapter Installing the covers

10.6.12 Skeleton replacement

To replace the skeleton, all components must be removed and the robotic hand reassembled from the beginning.

Perform in sequence:

1. Cover removal
2. Board removal
3. Tendon removal
4. Bowden tube removal
5. Spool removal
6. Finger removal
7. Servo motor removal, including the TRS Platform
8. SKELETON REPLACEMENT
9. Servo motor installation, including the TRS Platform
10. Finger installation
11. Spool installation
12. Bowden tube installation
13. Tendon installation
14. Board installation
15. Cover installation

10.7 Tendon registration

For each finger requiring registration:

1. **Secure the robotic hand in the vice**
2. connect the servo motor to the SSI slot on the electronic board.
3. power the electronic board with an appropriate power supply, see chapter 3 Machine overview
4. connect to the logic board via Bluetooth.
5. send the 'Tendon Installation Mode' selection serial command via Bluetooth.
6. **wait for the servo motors to complete their movement, then proceed to the next step**
7. For each finger requiring registration
    1. **Unscrew the extension tendon bolt by 2 turns using the 2mm hex key**
    2. **Unscrew the flexion tendon bolt by 2 turns using the 2mm hex key**
    3. Make sure the finger is fully extended; otherwise extend it manually
    4. Using the assembly pliers with one hand, take hold of the extension tendon and pull it until taut
    5. While keeping the tendon taut with the assembly pliers, with the other hand tighten the fastening bolt until firmly secured (TBD: tightening torque to be defined)
    6. Release the tendon from the assembly pliers
    7. Make sure the finger is fully extended under the effect of the tendon's tension
    8. Using the assembly pliers with one hand, take hold of the flexion tendon and pull it until taut

9. While keeping the tendon taut with the assembly pliers, with the other hand tighten the fastening bolt until firmly secured (TBD: tightening torque to be defined)
10. Release the tendon from the assembly pliers

11 Cleaning and disinfection

The external surfaces of the robotic hand may be cleaned with:

- a cloth lightly dampened with alcohol.
- a cloth lightly dampened with a chlorine-based disinfectant (e.g. Amuchina).



DO NOT SPRAY LIQUIDS ONTO THE ROBOTIC HAND

12 Fault detection / troubleshooting

FAULT	CAUSE
• Fingers limp/wobbly even when not moved by the motor • Fingers closing late • Fingers not closing fully	• tendons not taut (robotic hand used beyond load limits)
• Fingers not opening fully • Fingers not closing fully	• failed tendon calibration
• Fingers not moving	• damaged tendons • detached tendon • damaged motors • power failure • damaged spools
• Motors stop exerting force after a certain time	• motor overheating
• Cannot establish a Bluetooth connection with the robotic hand	• Distance to the controlling device too great, or interference, or obstacles
• Fingers not gripping the object	• Worn pads
• Damaged or worn parts	• Breakage or wear from use • Breakage or wear from improper use

CAUSE	SOLUTION
• Tendons not taut	• Remove the cover and perform tendon registration
• Damaged tendons	• Replace the tendons
• Failed tendon calibration	• Restart the robotic hand to repeat auto-calibration. • Power off the robotic hand, remove the covers, and check that the tendons have not become tangled with each other or with other parts of the robotic hand
• Detached tendons	• Power off the robotic hand, remove the covers, and reattach or replace the tendon, then perform tendon registration
• Motor overheating	• Make sure the robotic hand is used within the correct operating range
• Distance to the controlling device too great, or interference, or obstacles	• Bring the controlling device within Bluetooth range (<9m) • Remove obstacles between the controlling device and the robotic hand • Eliminate any interference
• Worn pads	• Replace the parts containing the damaged pads
• Power failure	• Check that the power supply is active and properly connected • Check that the power cables are not damaged
• Breakage or wear from use • Breakage or wear from improper use	• Replace damaged parts with new or manufacturer-reconditioned parts

13 Disassembly, deactivation and scrapping

13.1 Removing the robotic hand

Before removing the robotic hand, deactivate it by disconnecting the electrical power cable.

Perform in sequence to remove the robotic hand from the robotic manipulator:

- unscrew the 4 M3 bolts that hold the flange attached to the robotic hand
- detach the robotic hand from the flange
- unscrew the 4 M6 bolts that hold the flange attached to the manipulator's wrist
- detach the flange from the manipulator's wrist

13.2 Deactivating the robotic hand

To deactivate the robotic hand, disconnect the power cable.

13.3 Disassembly

Before disassembling the robotic hand, deactivate it and detach it from the machine on which it is installed.

To disassemble the robotic hand, perform in sequence:

- remove the robotic hand's covers
- remove the finger covers
- remove the palm
- remove the bolts from the spools
- remove the spools
- remove the tendons using the assembly pliers; cut them with scissors if necessary
- remove the fingers: index, middle, ring, little
- remove the flexible part of the thumb
- remove the thumb base
- remove the cable ties
- remove the tendon barrier
- remove the motors

13.4 **Scrapping**

At the end of its life, the product must not be disposed of as unsorted municipal waste.

Dispose of it in accordance with local regulations on waste electrical and electronic equipment.

As an alternative, the user may proceed to disassemble the product and dispose separately of the different types of components in accordance with the procedures indicated in the following paragraphs and in compliance with national and local waste regulations.

13.5 **Electronic components**

Electronic components (servo motors, cables, control board) must be disposed of in accordance with WEEE disposal regulations.



13.6 **Bolts, nuts and washers**

Bolts, nuts, washers, and other small metal parts, if separated from the other components, must be disposed of through separate collection systems dedicated to metals or mixed materials (where provided), in accordance with the procedures established by the local waste management system.

The user must comply with the specific regulations in force in their own country and with the instructions of the local waste management operator.

13.7 **Silicone pads**

The silicone-containing parts (flexible fingers, palm, and similar components) consist of already-cured silicone and do not contain substances known to be hazardous to the end user.

In the absence of dedicated collection channels, these components may be disposed of as unsorted municipal waste, in compliance with the specific regulations in force in the user's own country and with the instructions of the local waste management operator.

13.8 **Rigid plastic parts**

Rigid plastic parts (skeleton, thumb base, Bowden tubes, spools, covers, flange and similar components), if separated from the electronic and metal parts, must be disposed of through separate collection systems dedicated to plastics or mixed materials, or delivered to recycling centres, in

accordance with the local waste management system.The user must comply with national regulations and with the instructions of the local operator regarding the collection and recycling of plastic waste.

13.9 **Tendons**

The tendons are made of high-density polyethylene (HDPE), a dense plastic that must not be released into the environment.These components must be disposed of through separate collection systems for plastics, or delivered to recycling centres that accept non-hazardous plastic waste, in accordance with the local waste management system. The user must comply with applicable regulations and with the instructions of the local operator to ensure proper recovery or disposal of this material.



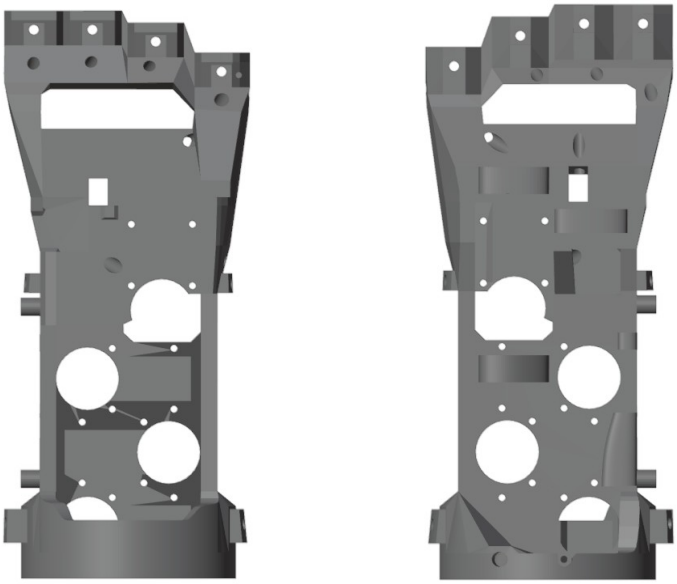
14 Visual documentation of the components

The following images include CAD renders for the designed components and photographs for the commercial components.

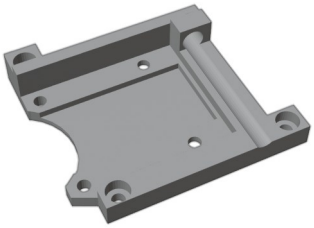
14.1 Components

For the technical specifications, quantities and codes of the spare parts, see chapter 10.3 Spare parts specifications.

14.1.1 Skeleton



14.1.2 TRS platform (Thumb Rotation Servo platform)



14.1.3 Thumb base



14.1.4 Bowden tubes

BOWDEN TUBES PHOTO TODO

14.1.5 Palm



14.1.6 Fingers

14.1.6.1 Index finger

The other fingers (middle, ring and little) are identical to the index finger.



14.1.6.2 Thumb



14.1.7 Servo motors and cables

TBD 5 x WaveShare ST3215HS



14.1.8 Logic board

TBD Waveshare ESP32 Servo Driver Board



14.1.9 Tendon spools





14.1.10 Polyethylene tendons

TBD Hercules brand 8-strand polyethylene, diameter 0.8 mm

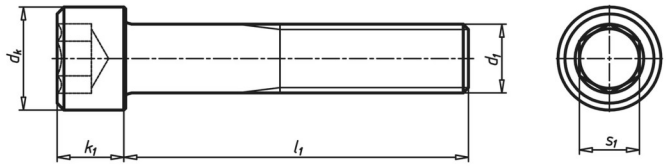


14.1.11 Screws and nuts

14.1.11.1 M2 cylindrical-head screws with hex socket

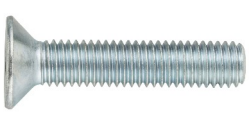


14.1.11.1.1 Length 5 mm



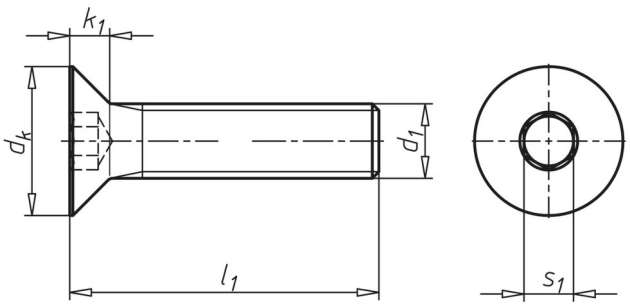
l1	d1	s1
5 mm	2 mm	1.5 mm

14.1.11.2 M3 countersunk-head screws with hex socket



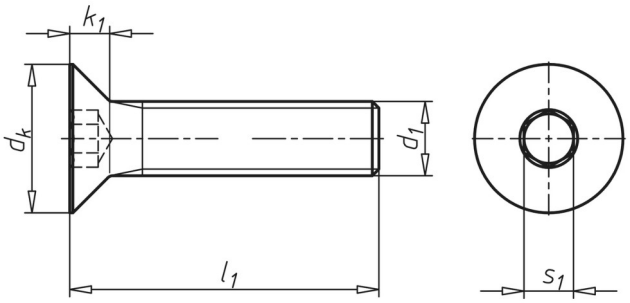


14.1.11.2.1 Length 20 mm



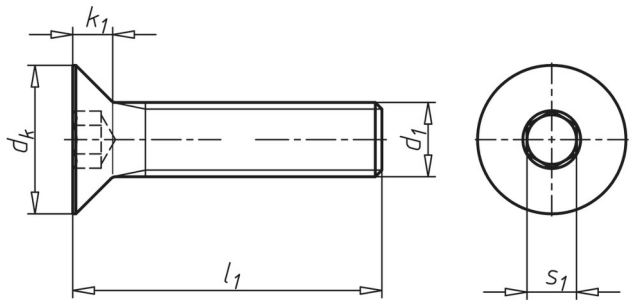
l1	d1	s1
20 mm	3 mm	2 mm

14.1.11.2.2 length 16mm



l1	d1	s1
16 mm	3 mm	2 mm

14.1.11.2.3 **length 12mm**



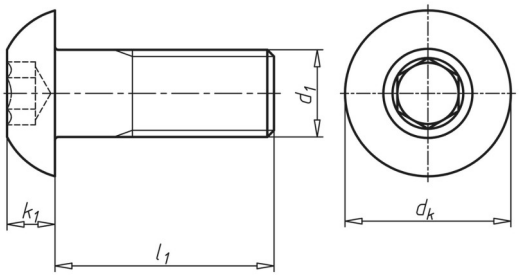
l1	d1	s1
12 mm	3 mm	2 mm

14.1.11.3**M3 button-head screws**

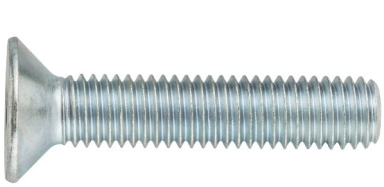


14.1.11.3.1 **length 10mm**

l1	d1	s1
10 mm	3 mm	2 mm

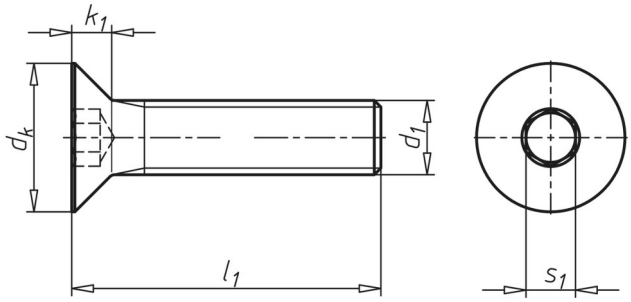


14.1.11.4 **M6 screws**





14.1.11.4.1 **Length 35 mm**

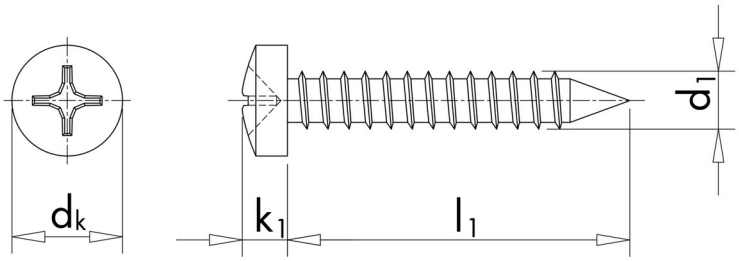


l1	d1	s1
35 mm	6 mm	4 mm

14.1.11.5 M2 self-tapping screws, button head, Phillips drive



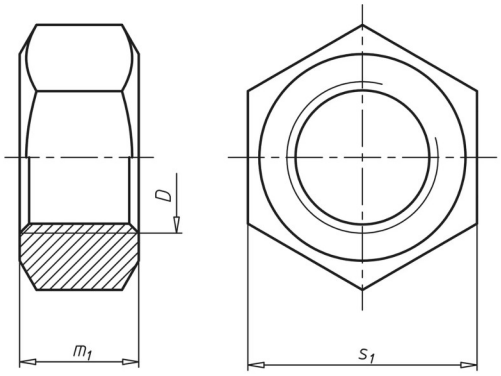
14.1.11.5.1 Length 6.5mm



l1	d1
6.5 mm	2 mm

14.1.11.6 **M3 hex nuts**

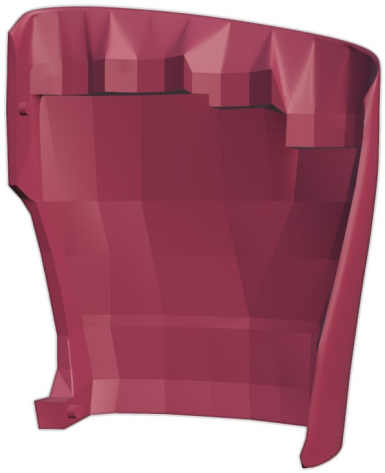




m1	s1
2.4 mm	5.5 mm

14.1.12 Covers

14.1.12.1 Rear upper cover



14.1.12.2 Front lower cover



14.1.12.3 Rear lower cover



14.1.12.4 Finger phalanx covers

14.1.12.4.1 Index finger covers

The covers of the other fingers (middle, ring and little) are identical to those of the index finger.



14.1.12.4.2 Thumb covers

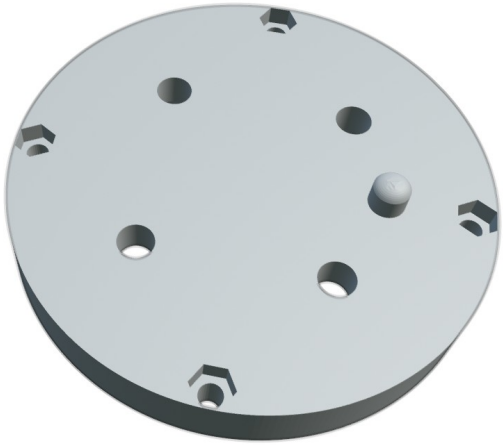


14.1.13 Mechanical connection flanges

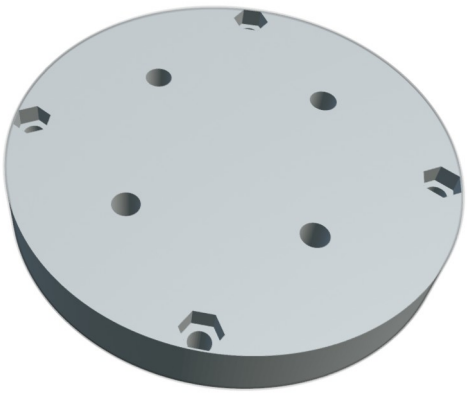
14.1.13.1 Flange ISO-9409-1-50-4-M6



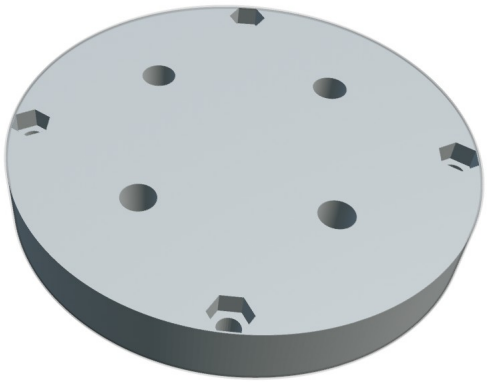
14.1.13.2 Flange ISO-9409-1-40-4-M6



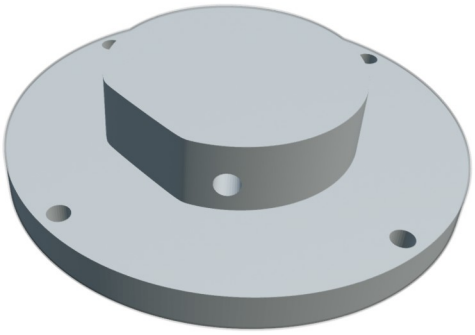
14.1.13.3 **Flange PCD-40-4-M5**



14.1.13.4 **Flange PCD-40-4-M6**



14.1.13.5 Flange for Agilex Black



14.2 **Hole and slot nomenclature**

Note: holes are indicated in red.

14.2.1 **Thumb servo motor hole nomenclature**

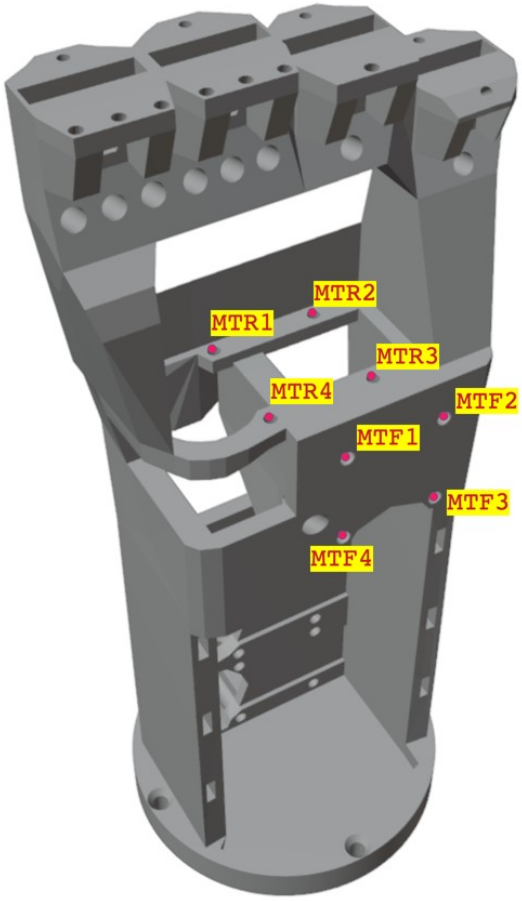


Figure 14.1: Left hand skeleton - front view - servo motor holes

14.2.2 Upper finger servo motor hole nomenclature

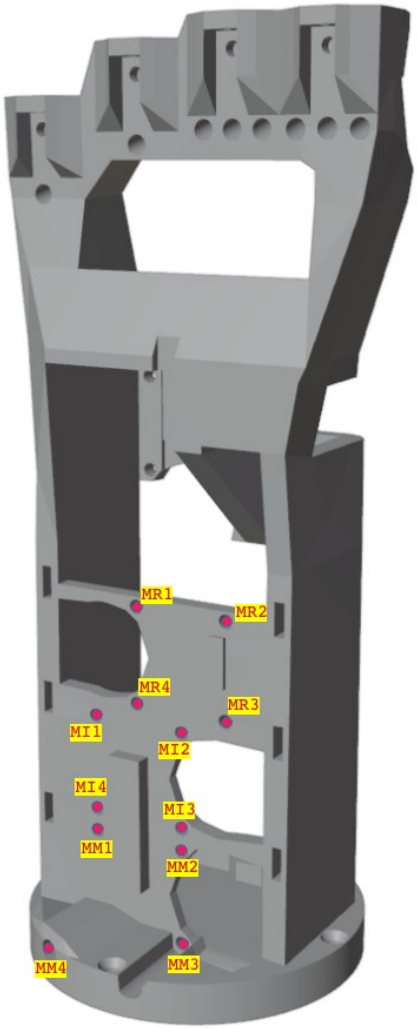


Figure 14.2: Left hand skeleton - rear view - servo motor holes

14.2.3 Servo motor slot nomenclature

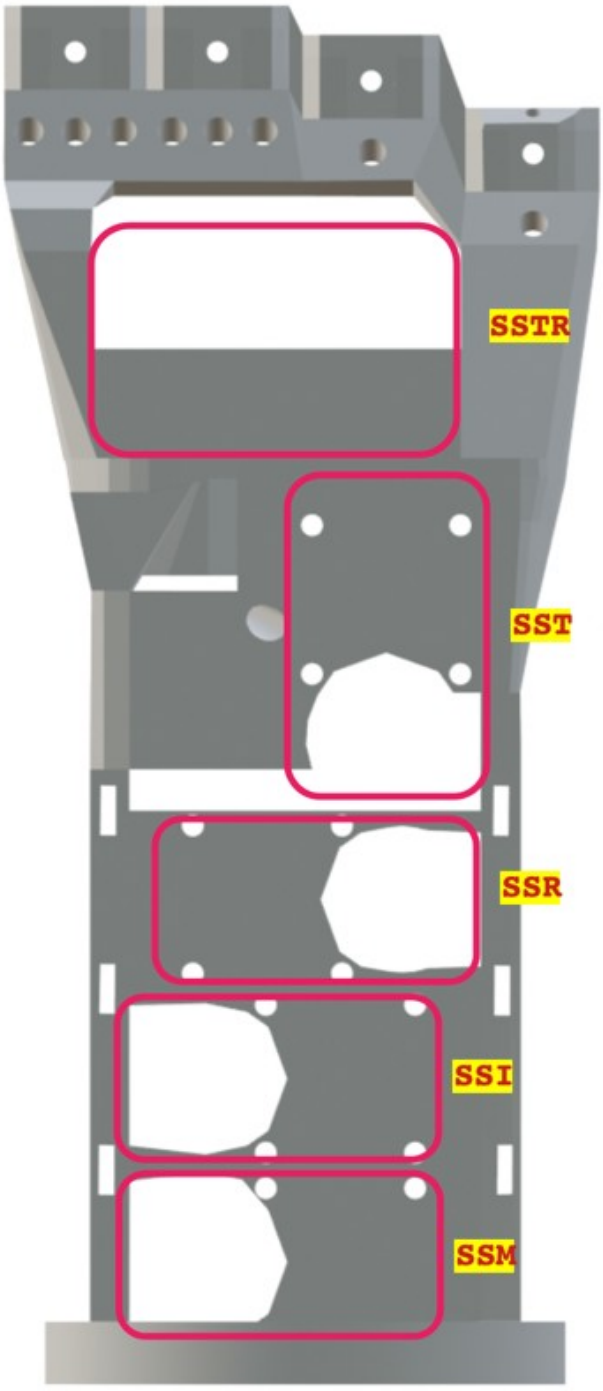


Figure 14.3: Left hand skeleton - front view - motor slots

14.2.4 Servo motor flange hole nomenclature



Figure 14.4: Waveshare ST3215 / ST3215-HS servo motor - holes

14.2.5 Upper finger slot and hole nomenclature

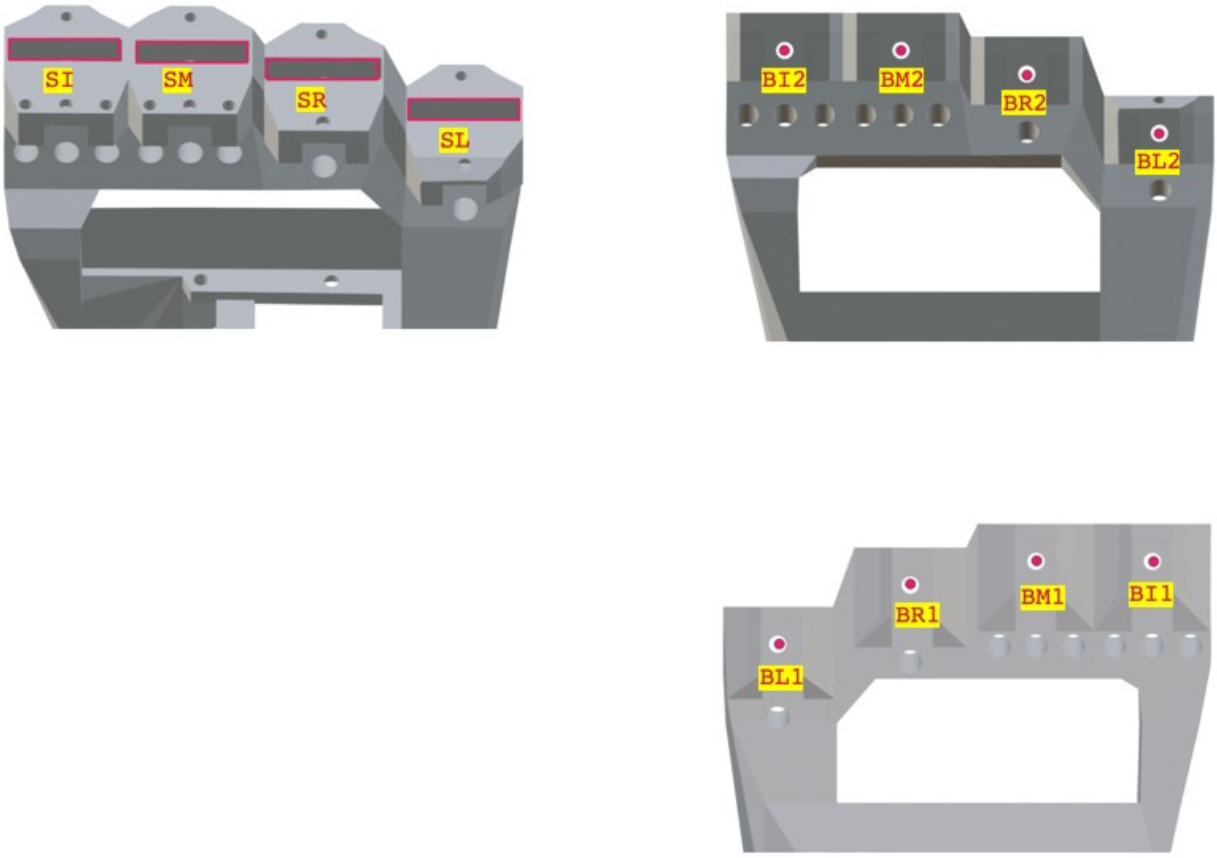


Figure 14.5: Left hand - skeleton - front view - upper finger holes and slots

14.2.6 Thumb base slot and hole nomenclature

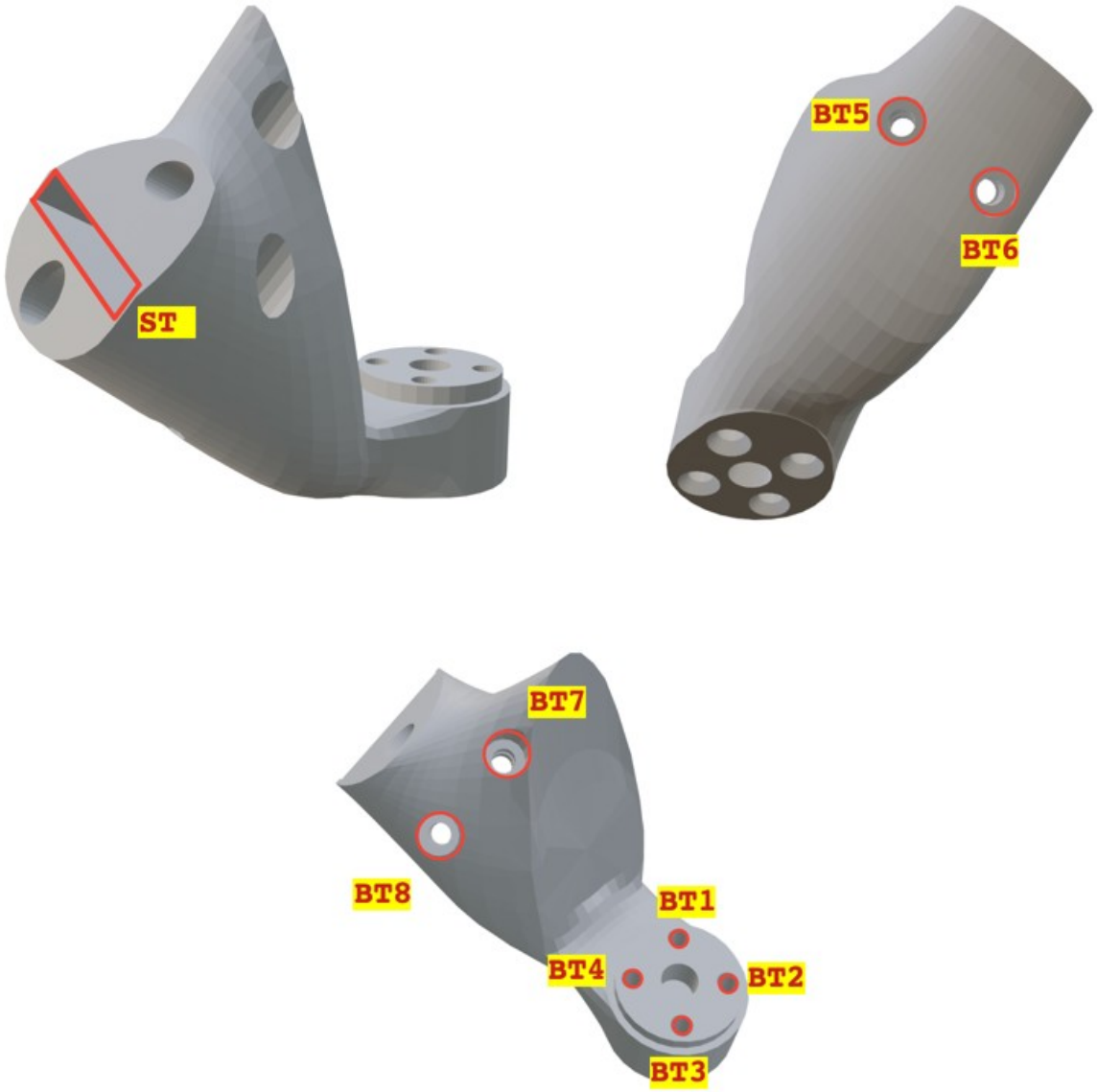


Figure 14.6: Left hand - front view - thumb base holes and slots

14.2.7 Tendon hole nomenclature

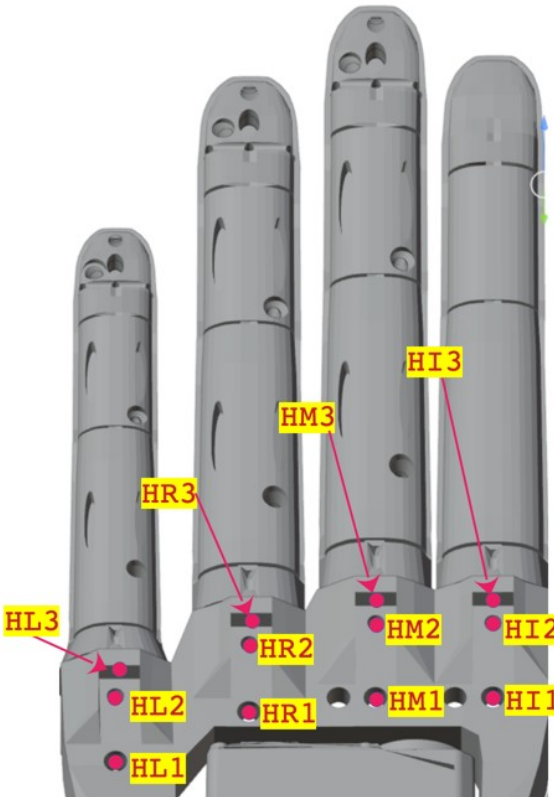


Figure 14.8: Left hand - rear view - tendon holes



Figure 14.7: Left hand - front view - tendon holes

Only the names for the index finger are shown. The hole names of the other fingers can be obtained by replacing the prefix HI with that of the desired finger. The prefixes for the names of the other upper fingers (middle, ring and little) are listed below.

- Middle: HM
- Ring: HR
- Little: HL



Figure 14.9: Index finger - distal tendon hole - view from below

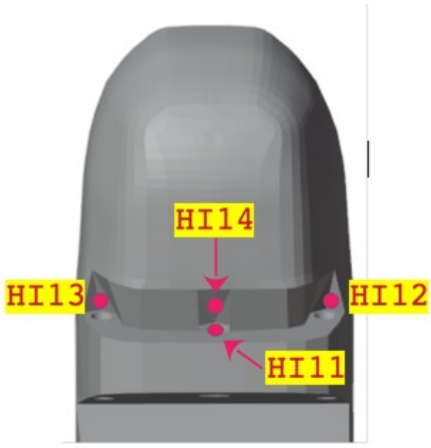


Figure 14.10: Index finger - distal tendon hole - view from above



Figure 14.11: Index finger tendon holes

14.2.8 Thumb tendon hole nomenclature



Figure 14.12: Thumb tendon holes

14.2.9 Spool hole nomenclature

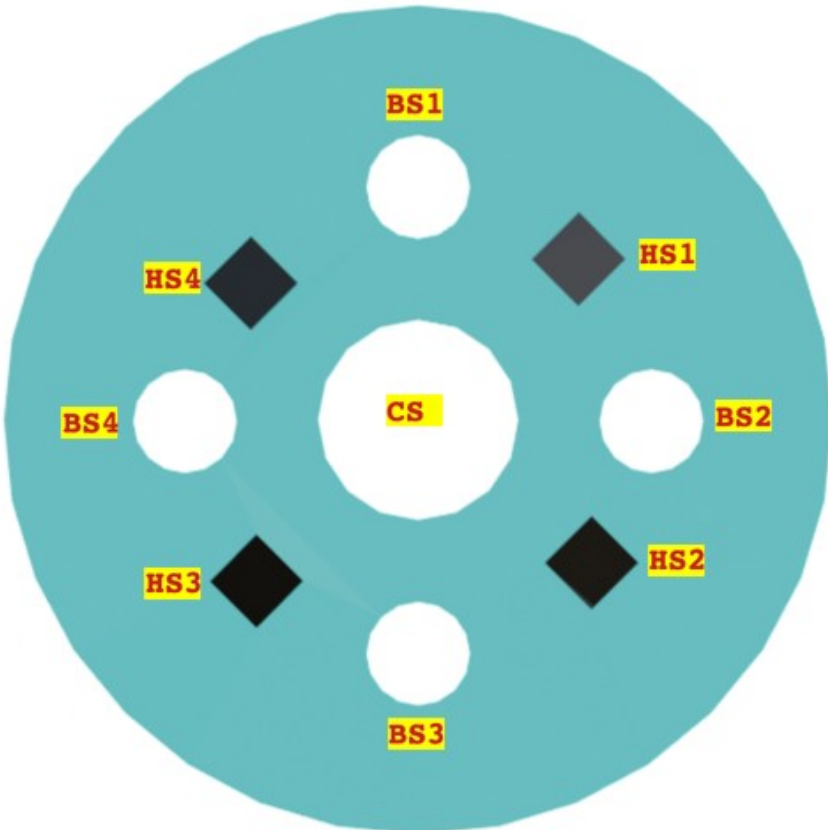


Figure 14.13: Spool holes

14.2.10 Palm hole nomenclature

TBD

14.2.11 Electronic board hole nomenclature

TBD

15 Residual risks to be communicated to users

15.1 General notices

This section contains the residual risks of the partly completed machinery that MUST BE COMMUNICATED TO END USERS in the manual of the complete machine. The integrator must incorporate this chapter in full.

15.2 Mechanical risks

- **Crushing risk: the robotic fingers exert force of up to 20N during grasping. Keep hands/objects away during operation.**
- **Exposed tendons: risk of snagging clothing/external objects. Do not touch the tendons during movement.**

15.3 Mandatory maintenance (normal wear)

Tendon tension adjustment (registration) is ordinary maintenance required due to normal wear from use, needed every 5000 cycles (one cycle = one finger closure to resistance, against the palm or an object) — similar to re-tensioning a bicycle brake cable. Failure to comply voids the warranty coverage. TBD The firmware automatically logs cycles; check the app for maintenance alerts.

15.4 Operating limitations

- **Do not operate standalone: the partly completed machinery is dangerous if not integrated into a complete robot equipped with emergency stop and safety barriers.**
- **Overload:**
  - Pinch grasp: 500 g **TBD**
  - Power grasp: 5 kg **TBD**
  - Lift: 8 kg **TBD**
- **Environment: Temperature 5-40°C, humidity <80% non-condensing.**

15.5 Operator training

Train operators on: the risks listed above, the tendon registration procedure, and the energy isolation procedure (Chapter 10.5)

The integrator must translate and include this chapter in the end-user manual of the complete machine.